

ADVANCING AI TRUSTWORTHINESS THROUGH PATIENT SIMULATION: RISK ASSESSMENT OF CONVERSATIONAL AGENTS FOR ANTIDEPRESSANT SELECTION

Md Tanvir Rouf Shawon*

George Mason University (GMU), Computer Science (CS), mshawon@gmu.edu

Mohammad Sabik Irbaz

GMU, Information Sciences and Technology (IST), mirbaz@gmu.edu

Hadeel R. A. Elyazori

GMU, IST, helyazor@gmu.edu

Keerti Reddy Resapu

GMU, Health Administration and Policy (HAP), kresapu@gmu.edu

Yili Lin

GMU, HAP, ylin26@gmu.edu

Vladimir Franzuela Cardenas

GMU, HAP, vcarden@gmu.edu

Farrokh Alemi

GMU, HAP

Kevin Lybarger

GMU, IST klybarga@gmu.edu

CCS CONCEPTS • Artificial intelligence • **Natural language generation** • **Natural language interfaces** • **Consumer health** • **Health informatics** •

Additional Keywords and Phrases: Patient simulation, conversational agents, AI risk management, trustworthy AI, large language models, depression management

* Corresponding author.

ABSTRACT

Objective: This paper introduces a patient simulator for scalable, automated evaluation of healthcare conversational agents, generating realistic, controllable interactions that systematically vary across medical, linguistic, and behavioral dimensions to support risk assessment across populations.

Methods: Grounded in the NIST AI Risk Management Framework, the simulator integrates three profile components: (1) medical profiles constructed from All of Us electronic health records using risk-ratio gating; (2) linguistic profiles modeling health literacy and condition-specific communication; and (3) behavioral profiles representing cooperative, distracted, and adversarial engagement. Profiles were evaluated against NIST AI RMF trustworthiness requirements and assessed against an AI Decision Aid for antidepressant selection.

Results: Across 500 simulated conversations, the simulator revealed monotonic degradation in AI Decision Aid performance across health literacy levels: Rank-1 concept retrieval ranged from 47.6% (limited) to 81.9% (proficient), with corresponding recommendation degradation. Medical concept fidelity was high (96.6% across 8,210 concepts), validated by human annotators (0.73 κ) and an LLM judge with comparable agreement (0.78 κ). Behavioral profiles were reliably distinguished (0.93 κ), and linguistic profiles showed moderate agreement (0.61 κ).

Conclusions: The simulator exposes measurable performance risks in conversational healthcare AI. Health literacy emerged as a primary risk factor with direct implications for equitable AI deployment.

1 INTRODUCTION

Conversational AI systems are increasingly deployed in healthcare to support clinical decision-making, patient engagement, and care accessibility [30, 39, 61, 62]. Yet these systems face a fundamental evaluation gap: patients communicate the same clinical information in vastly different ways depending on health literacy, psychological state, and interaction style [26, 69]. However, there is no standardized systematic method to assess whether AI performance remains equitable across this variation [22, 60]. A patient who says, “my morning pill for my nerves” and one who reports “20 mg of fluoxetine for generalized anxiety disorder” may describe identical clinical realities, but a conversational agent that handles the second better than the first introduces a structural bias against vulnerable populations. LLM-based approaches lower barriers to building such systems [42, 68] but heighten the need for rigorous, scalable risk assessment that can characterize performance across diverse patient communication patterns.

Static benchmarks cannot capture these risks because they bypass the multi-turn dynamics through which communication variation compounds [18, 27]. Patient simulation offers a scalable alternative: simulated patients with controlled clinical, linguistic and behavioral characteristics can systematically probe system performance across populations that would be difficult, slow, or ethically complex to recruit for live evaluation, enabling identification of vulnerabilities [4, 24]. The National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF) provides structured, domain-agnostic guidance for identifying, assessing, and managing AI risks [58], but no existing patient simulation framework operationalizes this guidance with the multidimensional variation needed to expose communication- and interaction dependent failures. This work introduces such a framework, grounding patient simulation in the AI RMF to enable empirical risk assessment of conversational healthcare AI.

We present a patient simulator grounded in real-world clinical data from the All of Us Research Program [3], evaluated on a conversational decision aid for antidepressant selection in major depressive disorder (MDD). The simulator integrates three profile dimensions: (1) medical profiles generated by a novel Medical Artificial General Intelligence (MAGI) algorithm that constructs outcome-relevant clinical histories from electronic health record (EHR) data through risk-ratio gating and independence screening; (2) linguistic profiles modeling health literacy variation [23, 43] and condition-specific

communication patterns [46, 50]; and (3) behavioral profiles representing empirically derived interaction patterns including cooperative, distracted, and adversarial engagement [49, 56]. While evaluated here on antidepressant selection, the framework generalizes across healthcare tasks and conditions. Building on prior work in patient simulation and NIST-aligned risk governance, this study addresses the following research questions:

RQ1. How effectively can structured EHR data be transformed into risk-aware medical profiles aligned with AI trustworthiness principles?

RQ2. How can simulated patients combining medical, linguistic, and behavioral information produce realistic and distinguishable conversational behavior?

RQ3. What performance risks and equity gaps does simulated patient evaluation reveal in a conversational clinical decision aid?

This work contributes: (1) a patient simulator integrating medical, linguistic, and behavioral profiles to operationalize the NIST AI RMF MAP and MEASURE functions for conversational healthcare AI, (2) the MAGI algorithm for generating outcome-relevant, auditable medical profiles from EHR data through risk-ratio gating and independence screening, (3) a controlled perturbation methodology using ontology-aware semantic substitution for validating both human annotators and LLM judges, and (4) empirical evidence that health literacy creates a monotonic performance gradient in conversational clinical AI, identifying a concrete equity risk. The code, source data, and simulated interactions are publicly available¹.

2 RELATED WORK

User simulation provides scalable methods for evaluating conversational agents across domains; patient simulation adapts these methods for clinical dialogue and diagnostic reasoning.

2.1 USER SIMULATION

User simulation enables controlled, scalable evaluation of interactive systems without requiring human trials. Foundational surveys span information access, dialogue modeling, and recommendation systems [4, 17, 41, 71], while recent work leverages LLMs to generate context-aware user behavior through dual-model architectures combining generators and verifiers [38, 52] and session-level search simulation [29, 76]. In healthcare, synthetic users with clinical profiles evaluate decision support and coaching systems [74], establishing user simulation as a critical method for assessing reliability and safety in AI systems.

2.2 PATIENT SIMULATION

Patient simulation involves construction of virtual patients that engage in interactive clinical dialogues with human or automated clinicians [13]. Frameworks vary widely in patient state representation, from hand-crafted profiles to EHR-grounded models, and in how they control disclosure, tone, and clinical accuracy [28, 30, 35]. Prior work often treats patient simulation as a single unified problem, but it involves two distinct technical challenges: (1) constructing clinically valid patient representations and (2) simulating interactive dialogue that expresses those representations over time [35]. Recent systems have integrated LLMs to enhance expressiveness and flexibility [21], constructed structured knowledge bases from clinical notes to support medical intake tasks [37], and embedded risk-aware feedback mechanisms [13].

¹ GitHub link to be provided upon acceptance.

2.2.1 Data-Driven Medical Realism

Early systems employed deterministic state machines or probabilistic sequence models to generate medical histories. Synthea [63] and SynSys [14] produce scalable, transparent timelines but reflect population-level distributions rather than individual patient data. EHR-grounded approaches improve realism. Generative Adversarial Network (GAN) variants [70, 72] address missing features and mixed data types, achieving higher fidelity than rule-based methods on MIMIC-III/IV benchmarks [11], while SimSUM [48] combines Bayesian networks with prompted LLMs to synthesize clinical notes. Systems integrating EHR grounding with conversational retrieval, including ophthalmology simulators [37] and multi-agent knowledge graph pipelines [73], advance clinical accuracy but treat the encounter as a transactional data exchange rather than a dynamic, rapport-dependent interaction.

2.2.2 Behavioral and Linguistic Modeling

Beyond medical profile construction, systems must express profiles through natural dialogue with appropriate communication style and conversational dynamics [8, 28]. Cognitive and persona-based approaches model psychological states to generate behaviorally coherent conversation. PATIENT Ψ [64] uses expert-designed cognitive schemas with GPT-4 to reproduce emotional fluctuations and resistant behaviors, while SFMSS [5] embeds Big Five traits to shape dialogue tone. These systems remain condition-specific (e.g., mental health) and lack medical grounding for clinical evaluation. Prompt-based approaches [13, 21] offer scalability but lack persistent patient state tracking and systematic control over linguistic and behavioral variation.

2.2.3 Procedural and Workflow Control

A third class prioritizes procedural fidelity and clinical workflows through discrete agent roles and actions. Bao et al. [5] coordinate patient, nurse, and supervisor agents for outpatient triage, while iPDG [66] enforces clinical plausibility through manually defined domain rules. These systems achieve procedural coherence but sacrifice conversational flexibility and behavioral depth.

2.2.4 Limitations and Gaps

Existing patient simulation systems exhibit two critical gaps that constrain trustworthy AI evaluation. First, no existing framework integrates medical realism, behavioral variation, and linguistic diversity within a unified architecture. Recent systems pair medical grounding with behavioral modeling or behavioral depth with linguistic variation, but this fragmentation creates evaluation blind spots. Existing systems cannot systematically test whether agents maintain diagnostic accuracy across complex comorbidities, varied health literacy, and adversarial behavior simultaneously. Multi-turn failures like context loss and inconsistent safety emerge precisely at these intersections.

Second, existing approaches prioritize medical outcomes over systematic risk management aligned with frameworks like the NIST AI RMF. They lack explicit risk mapping connecting simulation parameters to risk categories, auditability with traceable lineage to source data, and controllable risk probing through systematic variation. This prevents detection of consequential failures: agents may recommend different treatments when patients express identical conditions at different health literacy levels, or safety mechanisms may fail under adversarial conditions that only systematic variation can expose. We address both gaps through a unified framework whose medical, linguistic, and behavioral profiles operationalize the NIST AI RMF MAP and MEASURE functions, enabling risk assessment across medical accuracy, communication appropriateness, and behavioral robustness.

3 METHODS

3.1 Overview

The framework integrates a patient simulator with an *AI Decision Aid* to enable systematic evaluation of conversational clinical decision-making. The patient simulator produces controlled, profile-driven responses by combining medical, linguistic, and behavioral characteristics, while the *AI Decision Aid* conducts a structured conversational intake to elicit clinical history and generate antidepressant recommendations. Figure 1 illustrates this interaction. The remainder of this section details the patient simulator design, *AI Decision Aid* architecture, and evaluation methodology.

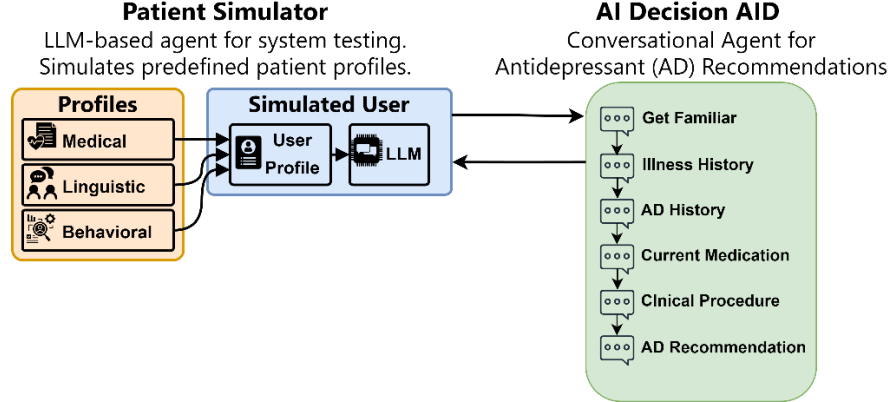


Figure 1: A schematic diagram of the conversation between patient simulator and the *AI Decision Aid* system.

3.2 Data

Both the patient simulator medical profiles and the *AI Decision Aid* prediction models are derived from the All of Us Research Program Registered Tier v8 dataset, a national longitudinal study providing structured EHR data including conditions, medications, procedures, and demographics [3].

Cohort Selection: We extracted data from All of Us participants with major depressive disorder (SNOMED CT Code 370143000 and its descendants). The resulting cohort includes 58,446 participants, contributing 466,752 antidepressant trials, with 18,471 diagnoses, 2,642 medications, and 5,001 procedures. Outcomes capture responses to 14 antidepressants and one category covering all remaining antidepressants (dataset cut-off: October 1, 2023). Antidepressant response was defined as taking the antidepressant for at least 10 weeks without switching or augmenting with another antidepressant [2].

Data Application: For patient simulator medical profiles, the dataset provides feature distributions, risk-ratio calculations for outcome-relevant predictor selection, and demographic distributions for age and gender initialization. The simulator was evaluated on the four most common antidepressants: fluoxetine, sertraline, trazodone, and duloxetine [79]. For the *AI Decision Aid*, the same data train prediction models that generate treatment recommendations across all 14 antidepressants. All data use complies with All of Us dissemination policies.

3.3 Patient Simulator Design

3.3.1 Design Rationale and NIST Alignment

The patient simulator design is grounded in the NIST AI RMF, which defines four core functions: (1) *GOVERN* – accountability and oversight, (2) *MAP* – risk identification, (3) *MEASURE* – risk assessment, and (4) *MANAGE* – risk

mitigation [58]. We operationalize MAP and MEASURE through three profile dimensions, each addressing a distinct category of risk: medical profiles targeting clinical accuracy, linguistic profiles targeting communication-dependent risks, and behavioral profiles targeting interaction-dependent risks.

Medical Profile Requirements. Medical profiles provide the clinical foundation for risk assessment, requiring realistic and diverse medical contexts to evaluate performance across clinical heterogeneity. Generating such profiles from structured EHR data faces interconnected challenges: high-dimensional feature spaces (10^5 - 10^6 concept codes) with sparse task-relevant signals, data quality issues (missingness, inconsistent coding), and medically implausible feature combinations [32, 53, 57]. Our design adheres to five AI RMF-aligned trustworthiness requirements: (1) *Controllability*: emphasize task-relevant features while maintaining outcome diversity. (2) *Coherence*: maintain clinical plausibility across diagnoses, treatments, and temporal events, including rare but valid scenarios. (3) *Variability*: capture heterogeneous comorbidities and contextual factors to expose brittleness and subgroup bias. (4) *Efficiency*: balance clinical completeness with computational tractability for large-scale evaluation. (5) *Transparency*: Maintain traceable feature lineage to source distributions for targeted risk analysis. Table 1 maps these requirements to data challenges and mitigated risks.

Table 1. Medical profile generation requirements: NIST alignment, challenges addressed, and risks mitigated.

Requirement (NIST Functions)	Data Challenge	AI Risk
Controllability (MAP, MEASURE)	High-dimensional feature space with sparse task-relevant signal	Spurious associations and overfitting; inflated performance estimates
Coherence (MAP, MEASURE)	Coding inconsistencies and implausible feature combinations	Medically contradictory profiles masking safety vulnerabilities
Variability (MAP, MEASURE)	Skewed distributions and underrepresentation	Subgroup disparities and incomplete risk detection
Efficiency (GOVERN, MEASURE)	Excessive record density limiting evaluation scale	Evaluation bottlenecks and reduced oversight capacity
Transparency (GOVERN, MEASURE)	Obscured feature provenance and latent interactions	Limited auditability and hidden bias propagation

Linguistic and Behavioral Profiles Requirements. Medical profiles establish clinical context but cannot capture communication-dependent and interaction-dependent risks, requiring two additional profile dimensions [34].

Communication-dependent risks emerge from heterogeneity in patient expression. Agents must maintain safety and comprehensibility across health literacy levels, condition-specific patterns, and vernacular variations [12, 40]. Linguistic profiles enable controlled assessment grounded in health literacy and psycholinguistic research [55].

Interaction-dependent risks emerge when patients manipulate information, test boundaries, or withhold details [77]. Because evaluation cannot assess safety mechanism robustness without systematic behavioral variation, behavioral profiles are grounded in clinical and human-computer interaction research to probe agent resilience across conditions.

Implementation. The patient simulator integrates three profiles: medical profiles that model clinical history and risk factors using probabilistic generation aligned with EHR data, linguistic profiles that capture variation in health literacy and condition-specific expression, and behavioral profiles that represent engagement and interaction patterns. Profiles are operationalized through structured LLM prompting with hierarchical indexing for traceability. Simulator-provided markers are removed before being passed to the *AI Decision Aid* to ensure independent extraction. Figure 2 shows an abbreviated prompt structure (complete version in Appendix 9.1.1).

Patient Simulator Prompt Structure (Abbreviated) You are simulating a patient based on three profiles: 1. Medical Profile (hierarchical indices for traceability): 1.1: Age: 34 2.1: Generalized Anxiety Disorder 3.2: Individual Psychotherapy [... additional diagnoses, medications, procedures ...] 2. Linguistic Profile: - Style: Concrete, informal - Tone: Hesitant, conversational [... vocabulary, structure specifications ...] 3. Behavioral Profile: - Adherence: High Engagement: High Focus: On-topic Adversarial: Minimal Response Generation: - Step 1: Identify relevant facts by index [e.g., 3.2] - Step 2: Apply linguistic style transfer - Step 3: Construct natural response with traceable references Example Output: <pre> "relevant_medical_history": ["[3.2] Individual Psychotherapy"] "style_transferred": ["[3.2] talked to someone"] "response": "<\s>talked to someone<\s> [3.2] for my anxiety" </pre>

Figure 2. Prompt Structure of Patient Simulator, abbreviated here for clarity. A more complete prompt is presented in Appendix 9.1.

3.3.2 Medical Profiles

Phase 1 generates patient profiles that are outcome-relevant, coherent, and diverse across clinical characteristics. Phase 2 applies probabilistic selection using a binomial distribution derived from All of Us antidepressant response data, ensuring the final cohort reflects realistic population-level variability.

Phase 1: Medical Profile Generation. Profile generation employs three core design principles [19, 20]: (1) prioritize outcome-relevant features, (2) enforce statistical independence among selected features, and (3) inject controlled diversity from residual features. These principles are implemented through a four-stage process: relevance filtering, demographic initialization, independence screening, and diversity expansion.

Stage 1: Top-K Filtering for Outcome Relevance. The algorithm restricts feature candidates to the top K predictors of the antidepressant response outcome, e_0 ($K = 500$). This relevance filter improves sample efficiency and focuses selection on features with demonstrated individual association to the target outcome aligning with the *max-relevance* component of minimum-redundancy-maximum-relevance (mRMR) feature selection [20].

Stage 2: Demographic Seeding. Each patient profile S is initialized with age and gender sampled from All of Us demographic distributions, grounding profiles in realistic population characteristics.

Stage 3: Independence-Screened Selection. Features are added iteratively, with each candidate v screened for statistical independence from already-selected features. The risk ratio $RR(s, v)$ quantifies association between features s and v with respect to antidepressant response:

$$RR(s, v) = \frac{P(response|s \cap v)}{P(response|s)}$$

where values near 1 indicate independence, values > 1.5 indicate positive association, and values < 0.67 indicate negative association. Each candidate must satisfy:

$$\frac{1}{1.5} < RR(s, v) \leq high, \forall s \in S$$

This symmetric band around unity excludes near-deterministic couplings ($RR > high$) and strong anticorrelations ($RR < 1/1.5$), ensuring statistically independent and clinically coherent feature combinations [20]. The upper threshold ($high$) was set to 7.

Stage 4: Diversity Expansion. After forming a coherent set of outcome-related features, we add up to a fixed number of residual features from outside the Top-K set, retaining only those with a meaningful association with at least one already selected feature:

$$RR(s, u) > 1.5, \exists s \in S$$

This captures latent contextual variables that enrich patient heterogeneity without compromising coherence, ensuring profiles reflect realistic complexity while remaining clinically plausible. This mechanism introduces bounded heterogeneity through correlated residual features, enriching diversity while maintaining clinical plausibility [19, 20]. Figure 3 shows an abbreviated algorithm: complete pseudocode and details are provided in Appendix 9.1.2.

Phase 2: Probabilistic Patient Selection. Phase 2 selects patients whose response probabilities collectively reproduce the population-level distribution observed in All of Us. Each Phase 1 patient has a response probability between 0 and 1. To ensure realistic population variability, the probability range is divided into seven sigma-bands of a binomial distribution $B(n, p)$, where n is the target cohort size and p is the population mean response rate. Medical profile sampling is weighted to match expected binomial frequencies. For example, with $n=100$ and $p=0.4$ ($\mu=40$, $\sigma \approx 4.9$), approximately 64% of patients fall within $(\mu - \sigma, \mu + \sigma]$, 15% in each adjacent band $(\mu \pm 1\sigma \text{ to } \mu \pm 2\sigma]$, and $<3\%$ in the tails beyond $\mu \pm 2\sigma$. This stratified sampling preserves central tendency and natural variability, while avoiding over- or under-representation of extreme responders. Detailed sigma-band allocations appear in Appendix 9.1.2.

AI RMF Alignment: *Controllability* is achieved through outcome-relevant feature selection with diversity across response probability levels. *Coherence* is ensured through risk-ratio screening, with the average risk ratio capturing internal clinical consistency. *Variability* is assessed via the number of unique concept codes across the profiles. *Efficiency* is quantified by the average number of features per profile, balancing completeness with computational tractability. *Transparency* is maintained through explicit feature lineage across all generation components, enabling auditable profile construction.

Algorithm: Medical Profile Generation

```

Input: Target outcome  $e_0$ , number of patients  $N$ , top- $K=500$ 

For each patient  $i = 1$  to  $N$ :

    Stage 1: Top-K Filtering for Outcome Relevance
     $V \leftarrow$  Top- $K$  predictors of antidepressant response  $e_0$ 

    Stage 2: Demographic Seeding Initialize
     $S \leftarrow \{\text{age, gender}\}$  sampled from population distributions

    Stage 3: Independence-Screened Selection
    For each candidate  $v$  in  $V$ :
        If  $\forall s \in S: 1/1.5 < RR(s,v) \leq \text{high}$ :
            Add  $v$  to  $S$ 

    Stage 4: Diversity Expansion
     $D \leftarrow$  number of expanded features
     $R \leftarrow$  residual features (outside top- $K$ )
    count  $\leftarrow 0$ 
    For candidate  $u$  in  $R$  while count  $< D$ :
        If  $\exists s \in S: RR(s,u) > 1.5$ :
            Add  $u$  to  $S$ 
            count  $\leftarrow$  count + 1

    Compute  $p \leftarrow$  predicted response probability for profile  $S$ 

Output:  $N$  patient profiles with associated response probabilities

```

Figure 3. Abbreviated medical profile generation algorithm. Complete algorithm details in Appendix 9.1.2.

3.3.3 Linguistic Profiles

Conversational agents must maintain safety and comprehensibility across diverse patient expression styles [10, 67]; systematic linguistic variation exposes blind spots and failure modes [31, 47]. The simulator implements a dual-axis linguistic framework with profiles along two independent dimensions: (1) a health literacy gradient capturing variation in comprehension, terminology, and discourse structure [9, 43, 44, 55] and (2) condition-specific communication reflecting linguistic patterns characteristic of depression and anxiety disorders derived from Linguistic Inquiry and Word Count (LIWC)-based clinical analyses [59]. Health literacy generalizes across clinical tasks as comprehension barriers affect patient-agent interaction regardless of medical condition. Condition-specific profiles capture diagnostic patterns (here depression and anxiety) adaptable to other conditions. Table 2 specifies the five linguistic profiles.

Table 2: Linguistic user profiles across multiple dimensions.

Profile	Key Linguistic Attributes	Example Response
Health Literacy		
Limited	<i>Style:</i> Concrete, informal, sometimes vague. <i>Tone:</i> Hesitant, uncertain, conversational. <i>Vocab:</i> Everyday terms, slang, vague quantities. <i>Structure:</i> Short, fragmented sentences; frequent fillers. <i>Patterns:</i> Minimal elaboration unless prompted.	<i>“Uh, just my morning pill... you know, the one for my nerves.”</i>
Functional	<i>Style:</i> Clear, basic descriptions of symptoms or routines. <i>Tone:</i> Cooperative, open. <i>Vocab:</i> Mix of common and medical terms. <i>Structure:</i> Simple narratives; occasional causal reasoning. <i>Patterns:</i> Provides coherent answers; asks clarifying questions.	<i>“I take Prozac every morning. It helps my mood, but I still have trouble sleeping.”</i>
Proficient	<i>Style:</i> Precise, clinical, well-organized. <i>Tone:</i> Confident, analytical. <i>Vocab:</i> Technical terms; qualifiers such as “likely” or “seems improved.” <i>Structure:</i> Multi-clause, logically sequenced sentences. <i>Patterns:</i> References timelines; anticipates follow-up questions.	<i>“I’m on fluoxetine, 20 milligrams daily. It’s effective, though I’ve noticed mild insomnia.”</i>
Condition-specific		
Depression	<i>Style:</i> Brief, muted, sometimes resigned. <i>Tone:</i> Flat, pessimistic, self-critical. <i>Vocab:</i> Negative emotion words; self-focused phrasing. <i>Structure:</i> Short, often past-tense statements. <i>Patterns:</i> Withdrawn responses; dismisses reassurance.	<i>“Barely sleeping. My head won’t shut off.”</i>
Illness Anxiety Disorder	<i>Style:</i> Symptom-focused and repetitive. <i>Tone:</i> Anxious, urgent. <i>Vocab:</i> Symptom terms; “what if” speculation; absolutist wording. <i>Structure:</i> Mix of run-on sentences and abrupt alarms. <i>Patterns:</i> Reassurance-seeking cycles; future-oriented worry.	<i>“I felt a flutter... what if it’s heart failure even though the test was normal?”</i>

AI RMF Alignment: This framework operationalizes Table 1 requirements through systematic, literature-grounded variation [9, 43, 44, 55, 59], enabling controllable assessment of communication-dependent risks.

3.3.4 Behavioral Profiles

Conversational agents struggle when patients go off-topic, respond vaguely, or withhold information [15, 16]. Systematic behavioral variation is essential for stress-testing agent robustness and assessing recovery from conversational breakdowns [33, 51]. We organized the 13 patient behaviors documented by Simpson et al. [54] into four behavioral categories (complete mapping in Appendix 10.1.3), with *Structured and Cooperative* as an additional fifth category. This study operationalizes three profiles: *Distracted & Unfocused* and *Adversarial & Combative* to capture challenging dynamics, with *Structured & Cooperative* as baseline. Each profile varies along four dimensions: conversational adherence, engagement, topical focus, and adversarial behavior, as detailed in Table 3.

Table 3: Behavioral user profile summary.

Profile	Key Attributes	Example Response
Structured & Cooperative	<i>Adherence: High</i> <i>Engagement: High</i> <i>Topical Focus: High</i> <i>Adversarial / Toxic Behavior: Minimal</i>	<i>“Yes, I take 20 mg of fluoxetine every morning around 8 AM. I haven’t missed a dose in the last three weeks.”</i>
Distracted & Unfocused	<i>Adherence: Low</i> <i>Engagement: Sporadic</i> <i>Topical Focus: Off-topic</i> <i>Adversarial / Toxic Behavior: Inadvertent derailment of conversation</i>	<i>“I was... wait, which one? Oh right, yeah I think? But yesterday I forgot — also my dog wouldn’t eat.”</i>
Adversarial & Combative	<i>Adherence: Variable</i> <i>Engagement: Variable</i> <i>Topical Focus: Variable</i> <i>Adversarial / Toxic Behavior: Overtly confrontational or hostile</i>	<i>“What kind of dumb question is that? Maybe if your system worked better, I wouldn’t have to answer this again.”</i>

AI RMF Alignment: Operationalizes Table 1 requirements through empirically grounded behavioral variation [54] enabling controlled assessment of interaction-dependent risks including adversarial scenarios with transparent lineage.

Together, these three profile dimensions provide comprehensive MAP and MEASURE coverage for conversational healthcare AI risk assessment.

3.4 CoT Prompting Strategy

We employ Chain-of-thought (CoT) prompting [65] to generate realistic patient responses through a structured process. For each question posed by the *AI Decision Aid*, the prompt directs the model to (1) identify relevant indexed medical attributes, (2) apply controlled term-level linguistic transformations, and (3) construct a natural language response with explicit references. Behavioral constraints are enforced as rules governing interaction, while linguistic constraints shape the form of expression. All outputs use a fixed JSON schema, enabling traceability and systematic evaluation (details in Appendix 9.1.1).

3.5 AI Decision Aid

The *AI Decision Aid* is a multi-agent conversational platform for antidepressant selection, serving as the system under evaluation in this black-box assessment. The system conducts structured intake through six sequential stages: establishing rapport, collecting illness history, gathering antidepressant history, documenting current medications, recording clinical procedures, and generating personalized recommendations. Each stage employs LLM-guided dialogue to elicit clinical information and a Retrieval-Augmented Generation (RAG) system to normalize medical concepts into an analytical reasoning system for estimating antidepressant response. The analytical advice system is based on the Medical Artificial General Intelligence Algorithm (MAGI) [Citation removed to preserve anonymity], which estimates dependent Bayesian relationships among medical concepts to support clinical reasoning under partial observability. Alternative recommendation models (e.g. logistic regression) are compatible. System specifications and code are publicly available².

² GitHub link to be provided upon acceptance.

3.6 Evaluation Framework

The evaluation framework examines patient simulator performance across medical, linguistic, and behavioral dimensions. Medical profiles are assessed through human and LLM-based annotation applying a three-label schema (*Accurate*, *Inaccurate*, *Unsupported*). Linguistic and behavioral profiles are evaluated through human annotation, quantitative metrics, and visual clustering to assess profile distinctiveness. This multi-method approach validates both simulator compliance with specified characteristics and clinical realism while maintaining interpretability and scalability.

3.6.1 Human Annotation

All annotations were performed by two annotators with complementary domain expertise: one with a graduate degree in Psychology and one a licensed Nurse Practitioner. Each sample was doubly annotated, with disagreements resolved through adjudication.

3.6.2 Medical Profile Evaluation

The patient simulator CoT process: (1) identifies relevant medical concepts, (2) rephrases each concept according to the linguistic profile, and (3) constructs a natural language response based on linguistic and behavioral characteristics. For each conversational turn, annotators assess whether medical concepts identified by the simulator as relevant are accurately expressed in the final response. For each conversational turn, the simulator outputs (a) the set of medical concepts identified as relevant to the current question, referenced numerically (e.g., [2,3]), and (b) a natural language response with inline spans marking where each concept is expressed. These spans serve as annotation units: annotators label each span as *Accurate* or *Inaccurate* based on whether the underlying medical fact is faithfully expressed. Annotators additionally flag any medical content appearing outside tagged spans that does not correspond to the patient's profile as *Unsupported*, capturing hallucinated or fabricated clinical details. Table 4 summarizes the labels. Minor colloquialisms (e.g., "happy pills" for "Prozac") are acceptable only if core meaning is preserved. *Unsupported* concepts appear in the responses without corresponding references or indicating spans. This approach evaluates expression fidelity conditioned on correct concept retrieval. Concept recall, the proportion of patient profile concepts expressed during the conversation, is computed programmatically from the simulator's traced outputs, with 95% of concepts in the medical profile appearing at least once in the generated conversations.

Table 4. Labels for evaluating the medical profile in patient simulator

Label	Definition
Accurate	Medical fact present in the reference medical history and correctly expressed with realistic phrasing. Minor conversational simplifications (e.g., "happy pills" for "Prozac") permitted if core clinical meaning is preserved.
Inaccurate	Medical fact present in the reference medical history but misrepresented, distorting critical clinical details or using implausible phrasing.
Unsupported	Medical fact not present in the reference medical history; fabricated, speculative, or unrelated to the provided profile.

Controlled Error Injection for System Validation. The patient simulator expresses nearly all medical concepts retrieved by the CoT approach accurately, so annotators could achieve artificially high inter-annotator agreement (IAA) by labeling all concepts as *Accurate*, and the LLM judge validation would lack discriminative power. To enable rigorous evaluation, we introduce controlled semantic perturbations by replacing clinical concepts with semantically similar but clinically distinct alternatives. For example, "hypertension" might be replaced with "prehypertension" or "diabetes

mellitus" with "prediabetes," conditions that are semantically related but sufficiently clinically distinct to detect as *Inaccurate*.

Perturbations are selected through a two-stage process. First, semantic search over a vector database of SNOMED CT and CPT4 concepts identifies the top 20 semantically similar candidates. Second, candidates are randomly shuffled before ontology-based filtering removes hierarchical ancestors or descendants and enforces minimum hierarchical distance: the first qualifying candidate is selected, ensuring perturbations vary rather than defaulting to the nearest semantic neighbor. If no candidate qualifies, the similarity threshold is relaxed or the candidate pool is expanded. The resulting perturbed profiles maintain linguistic realism while introducing subtle clinical inaccuracies for evaluating both human annotation quality and LLM judge performance.

3.6.3 Linguistic Profile Evaluation

Linguistic profiles are evaluated through human annotation, where annotators classify each conversation into one of five predefined linguistic profiles. Five automated metrics are also used:

- 1. Reading Level:** Flesch–Kincaid Grade Level (FKGL) [25] estimates school grade level and is calculated per response turn and averaged per conversation.
- 2. Average Response Length:** Average words per turn using NLTK’s English tokenizer [36].
- 3. Medical Term Density:** Clinical term count via greedy n-gram matching (up to 6-grams) against the study medical lexicon (all concept codes present in the database).
- 4. Depression Score:** Mean turn-level probability from an XLM-RoBERTa-based depressive symptom classifier [78].
- 5. t-SNE:** Projects linguistic features to two dimensions using cosine distance colored by profile. Multiple random seeds tested to identify the run with lowest KL divergence.

3.6.4 Behavioral Profile Evaluation

Annotators classify each conversation into one of three behavior profiles. Three automated metrics quantify behavioral patterns:

- 1. On-topic similarity:** Average cosine similarity between each *AI Decision Aid* turn and the patient simulator response.
- 2. Toxicity:** Mean turn-level probability from the XLM-RoBERTa-based toxicity classifier [6].
- 3. t-SNE:** Analogous to linguistic t-SNE, projecting behavioral features and color-coding by behavioral profile.

3.6.5 LLM Judge Evaluation

LLM-based judges have demonstrated strong performance in healthcare evaluation tasks [7, 45, 75]. We employ a separate LLM, Claude Opus 4.6, as an automated judge, applying the same annotation schema across medical, linguistic, and behavioral dimensions. The judge prompt was developed on 45 held-out conversations without human annotations. Judge alignment was then validated against human annotations on the full evaluation set, with results in Section 4.

3.7 Experimental Paradigm

Experiments used 60 medical profiles evaluated across four antidepressants (Sertraline 17%, Trazodone 15%, Fluoxetine 13%, and Duloxetine 11%) selected to reflect common clinical practice [79]. For each antidepressant, we simulated a pool of patient profiles and down-sampled to 15 profiles per drug by stratifying across response probability bands. With *high* = 7 and 3-5 residual features per medical profile, generation yielded an average of 8-9 clinical features

per profile, consistent with reports that outpatient visits address 5.4 -7.1 clinical items per encounter [1]. This feature density reflects plausible patient disclosure and contrasts with All of Us records (approximately 98.88 features per patient).

The design comprises three evaluation settings: (1) linguistic variation – five linguistic profiles with fixed *Structured & Cooperative* behavior (300 conversations), (2) behavioral variation – three behavioral profiles with fixed *Functional Health Literacy* (180 conversations), and (3) intersectional analysis – combined linguistic-behavioral variation across a representative subset (150 conversations). After accounting for shared profiles across settings, the design yields 500 unique simulated conversations. The patient simulator and *AI Decision Aid* are powered by GPT-4.1, while Claude Opus 4.6 is used as the LLM-based judge. All embedding-based analyses, including linguistic and behavioral t-SNE projections and on-topic similarity, use OpenAI's *text-embedding-3-small* model, which also supports RAG-based concept normalization in the *AI Decision Aid*. The conversational *AI Decision Aid* and patient simulator are implemented as separate agents within LangGraph³.

4 RESULTS

4.1 Simulated Medical Profile Validation

Across 60 medical profiles, the framework demonstrates systematic risk characterization aligned with the NIST AI RMF MAP and MEASURE functions. *Controllability* was achieved by emphasizing outcome-related features comprising 37.16% of features per profile, while maintaining outcome diversity across response probability bands for each antidepressant. *Coherence* was reflected in an average risk ratio of 2.64 among outcome-related features, consistent with the risk-ratio gating. *Variability* was maintained through concept diversity, with profiles containing 292 unique concept codes across the 60 profiles. *Efficiency* was demonstrated by an average of 8.08 features per profile, substantially lower than the 98.88 features in a typical All of Us record. *Transparency* was maintained by explicitly preserving feature provenance and distributional fidelity across all generation components: demographic attributes followed All of Us distributions (ages 13-19: 0.14%, 20-40: 25.32%, 41-64: 41.46%, 65-79: 27.95%, 80-89: 5.13%; gender: male: 37.33%, female: 62.67%); outcome relevance was enforced using the Top-K set; residual features were drawn from the non-Top-K pool; and outcome probabilities matched antidepressant-specific binomial distributions derived from All of Us: fluoxetine (0.28-0.57, mean 0.41), sertraline (0.30-0.59, mean 0.44), duloxetine (0.28-0.57, mean 0.43), and trazodone (0.16-0.42, mean 0.29).

4.2 Medical Profile Validation

Medical profile fidelity was assessed through 1,786 simulator-generated medical concepts with annotated spans across 100 conversations, including 292 perturbed concepts. Table 5 shows IAA between human annotators and the LLM judge. Across all concepts, human IAA was high (0.73 κ , 0.93 F1). Agreement decreased for perturbed concepts (0.30 κ , 0.76 F1), reflecting the difficulty of subtle semantic substitutions. Human-LLM judge agreement followed the same pattern: 0.78 κ , 0.93 F1 overall; 0.24 κ , 0.77 F1 on perturbed concepts. Annotators identified 109 unique *Unsupported* cases, predominantly incidental medication mentions (e.g., Tylenol, ibuprofen, vitamins) and minor symptom references (e.g., headache, neck pain) not present in the medical profiles (IAA of 0.75 F1). The LLM judge identified 76 *Unsupported* cases of the same type (0.58 F1 against adjudicated human annotations). Individual annotator-LLM agreement (0.70-0.71 κ across all 1,786 concepts) did not differ significantly from human-human agreement by paired bootstrap ($p = 0.21$, $p =$

³ <https://github.com/langchain-ai>

0.32), nor did LLM agreement differ between the two annotators ($p = 0.63$). LLM-human agreement was comparable to human-human agreement.

Table 5. Agreement metrics for unperturbed and perturbed concepts

Evaluation	Concepts Set	n	Cohen’s κ	Accurate F1	Inaccurate F1	Unsupported F1	Overall F1
Human-Human	Perturbed	292	0.30	0.45	0.85	-	0.76
	Full	1786	0.73	0.96	0.76	0.75	0.93
Human-LLM	Perturbed	292	0.24	0.37	0.87		0.77
	Full	1786	0.78	0.97	0.81	0.58	0.93

LLM Judge Evaluation of Medical Profiles. Across 8,210 expressed concepts, 96.61% were labeled *Accurate*, 1.07% *Inaccurate*, and 2.31% *Unsupported*. Table 6 shows medical profile fidelity across linguistic profiles under fixed *Structured & Cooperative* behavior. The simulator maintained high accuracy (96-99%) and low error rates across all profiles. Within this range, concept expression frequency increases across the literacy gradient, peaking with *Proficient*. Error types diverge modestly by profile: *Proficient* shows the highest *Unsupported* rate (1.0%) due to greater medical term density, while *Illness Anxiety Disorder* produces the highest inaccuracy rate (2.5%) due to its repetitive tone. The *Depression* profile produces the fewest expressed concepts with minimal errors.

Table 6: LLM Judge Evaluation of Medical profile across the linguistic profiles under *Structured & Cooperative* behavior

Profile Name \ Evaluation Metrics	Total	Accurate	Inaccurate	Unsupported
Health Literacy	Limited	863	98.4%	1.3%
	Functional	952	98.2%	1.4%
	Proficient	981	98.7%	0.3%
Condition-specific	Depression	842	98.1%	1.1%
	Illness Anxiety Disorder	1,192	96.1%	2.5%
TOTAL	4,830	97.8%	1.4%	0.8%

Table 7 reveals that behavioral variation drives reliability shifts under *Functional Health Literacy* profile. *Structured & Cooperative* ensures stability, while *Distracted & Unfocused* behavior introduces frequent *Unsupported* outputs. *Adversarial & Combative* minimizes errors, likely because confrontational responses tend to be terse and narrowly responsive, limiting opportunities for *Unsupported* or *Inaccurate* content.

Table 7: LLM Judge Evaluation of Medical Profile across the behavioral profiles under a *Functional Health Literacy* linguistics

Behavioral Profile	Total	Accurate	Inaccurate	Unsupported
Structured & Cooperative	952	98.2%	1.4%	0.4%
Distracted & Unfocused	988	89.5%	0.5%	10.0%
Adversarial & Combative	1,095	99.3%	0.0%	0.7%
TOTAL	3,035	95.7%	0.6%	3.7%

Figure 4 shows that the intersection of linguistic and behavioral profiles shapes the *type* of error rather than overall accuracy, which remains high across all combinations. *Distracted & Unfocused* behavior produces elevated *Unsupported* rates concentrated in health literacy profiles, particularly *Limited* and *Functional*, where conversational drift introduces content outside the medical profile. *Adversarial & Combative* behavior drives inaccuracies, concentrated in *Proficient* and *Illness Anxiety* profiles where confrontational responses distort rather than fabricate clinical content. *Structured & Cooperative* behavior yields the lowest error rates overall, though *Illness Anxiety* produces *Unsupported* content even under cooperative conditions.

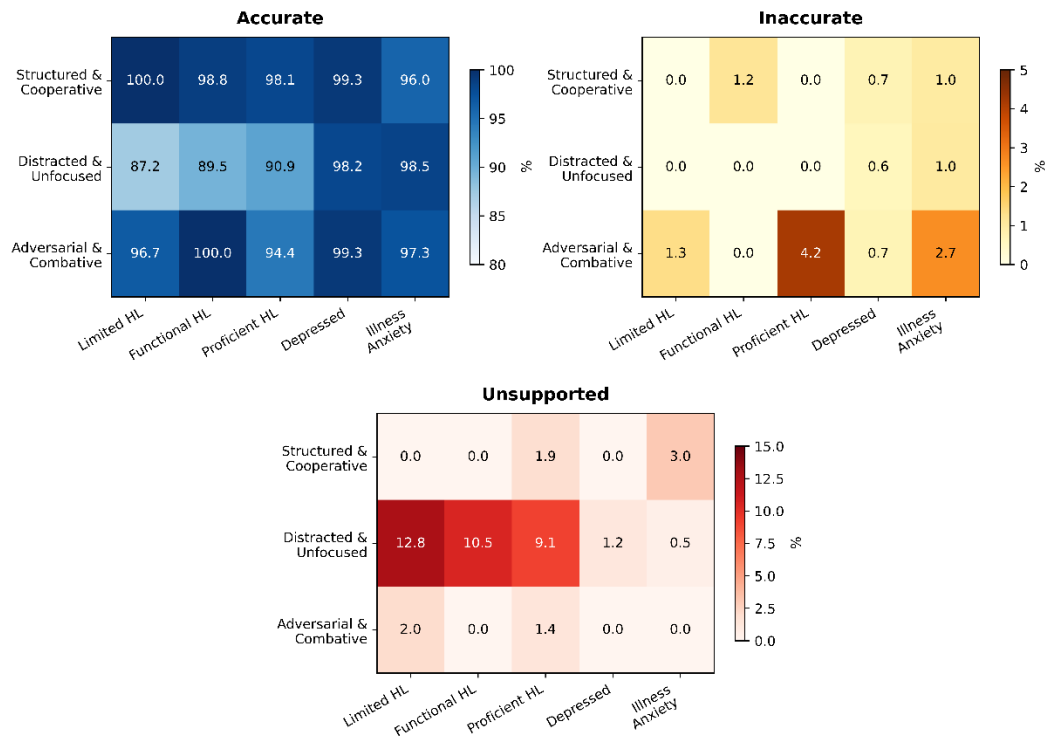


Figure 4: Results from the intersection of Linguistic and Behavioral Profiles

4.3 Linguistic Profile Validation

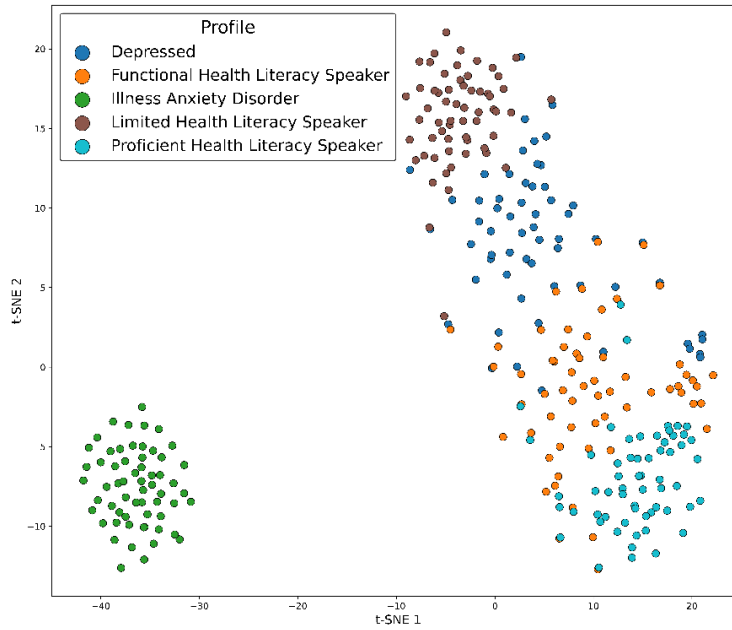
Human annotators showed moderate agreement on linguistic profile classification (0.61 κ , 0.70 micro F1). Human-LLM agreement was comparable (0.63 κ , 0.70 micro F1) in linguistic profile classification. When adjudicated labels were compared to the simulator's predefined profiles, classification reached 0.87 micro F1, indicating consistent expression of intended linguistic profiles. The LLM judge achieved 0.95 micro F1 across 500 conversations.

Table 8 shows linguistic variation under fixed *Structured & Cooperative* behavior. Reading level increases monotonically across the literacy gradient from *Limited* (3.59 FKGL) to *Proficient* (11.90 FKGL). Response length and medical term density increase across the health literacy gradient (*Limited*: 34.34 words, 4.33 medical terms; *Proficient*: 38.91 words, 11.68 medical terms). The *Depression* profile produces the shortest responses (21.48 words) with a comparable depression score (0.20), whereas *Illness Anxiety Disorder* produces the longest responses (65.05 words) with the highest medical term density (14.27) and a similar depression score (0.25), reflecting greater health-related elaboration.

Table 8 Evaluation of linguistic profiles under a *Structured & Cooperative* behavioral condition.

Profile Name \ Evaluation Metrics		Reading Level (FKGL)	Response Length	Medical Term Density	Depression Score
Health Literacy	Limited	3.59	34.34	4.33	0.01
	Functional	7.63	28.70	9.83	0.00
	Proficient	11.90	38.91	11.68	0.00
Condition-specific	Depression	4.49	21.48	6.17	0.20
	Illness Anxiety Disorder	8.11	65.05	14.27	0.25

Figure 5 shows t-SNE clustering where linguistic profiles form distinct clusters. Overlap between *Functional* and *Proficient Health Literacy* profiles reflect their shared characteristics. The *Depression* profile overlaps partially with lower literacy profiles, while *Illness Anxiety Disorder* forms a distinct cluster. Together, agreement scores, quantitative metrics, and embedding separation confirm that the simulator produces graded, distinct linguistic profiles.

Figure 5: t-SNE visualization of response embeddings for linguistic profiles under a *Structured & Cooperative* behavioral condition.

4.4 Behavioral Profile Validation

Human annotators showed high agreement on behavioral profile classification (0.93 κ , 0.96 micro F1). Human-LLM agreement was identical (0.93 κ , 0.96 micro F1). Classification against predefined profiles reached 0.98 micro F1. Table 9 shows behavioral variation under fixed *Functional Health Literacy*. *Structured & Cooperative* achieves baseline on-topic similarity (0.51) and the lowest toxicity (0.0003). *Distraacted & Unfocused* shows slightly reduced on-topic similarity (0.50), consistent with conversational drift, while maintaining low toxicity (0.0035). *Adversarial & Combative* exhibits substantially higher toxicity (0.0564) while retaining the highest topical relevance (0.52).

Table 9. Evaluation of behavioral profiles under a *Functional Health Literacy* linguistic condition.

Profile	On-topic Similarity	Toxicity
Structured & Cooperative	0.51	0.0003
Distracted & Unfocused	0.50	0.0035
Adversarial & Combative	0.52	0.0564

Figure 6 shows t-SNE clustering with clear separation among behavioral profiles. Agreement scores, quantitative metrics, and embedding separation confirm distinct behavioral profiles under fixed linguistic conditions.

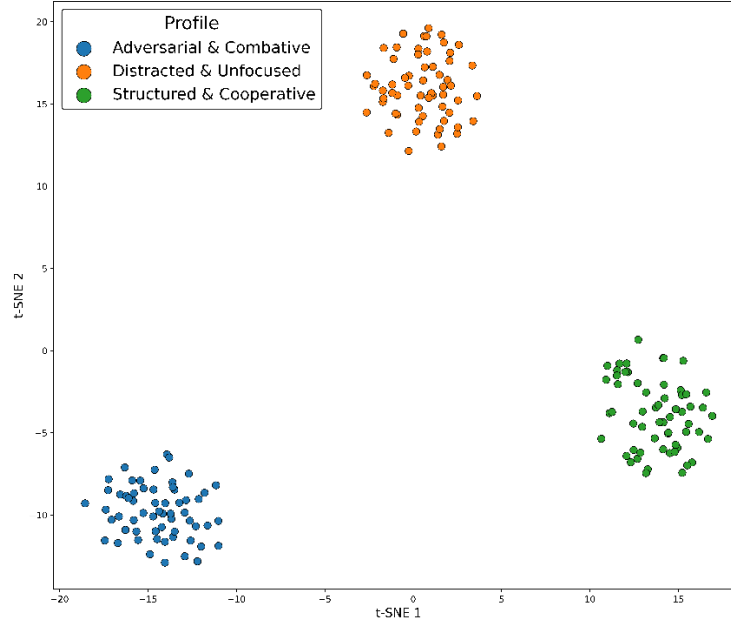


Figure 6: t-SNE visualization of response embeddings showing clear separation among behavioral profiles under a fixed *Functional Health Literacy* condition.

4.5 Profile Interactions

Figure 7 presents intersectional evaluation across linguistic and behavioral profile combinations. Key interaction patterns emerge: *Distracted & Unfocused* producing the longest responses across all linguistic profiles (61-82 words vs. 21-66 for *Structured & Cooperative*). *Depression* profiles consistently produce the shortest responses across behaviors (21-32 words) while maintaining elevated depression scores (0.18-0.25). Toxicity shows the strongest behavioral dominance, with *Adversarial & Combative* profiles exhibiting elevated toxicity (0.03-0.11) regardless of linguistic profile. Reading level and depression scores remain largely determined by linguistic profiles, with minimal behavioral modulation. On-topic similarity shows modest behavioral effects, with *Structured & Cooperative* achieving slightly higher alignment than *Distracted & Unfocused*. These patterns demonstrate that the simulator maintains profile independence where intended (linguistic features preserved across behaviors) while capturing realistic interactions (behavioral effects on engagement and tone).

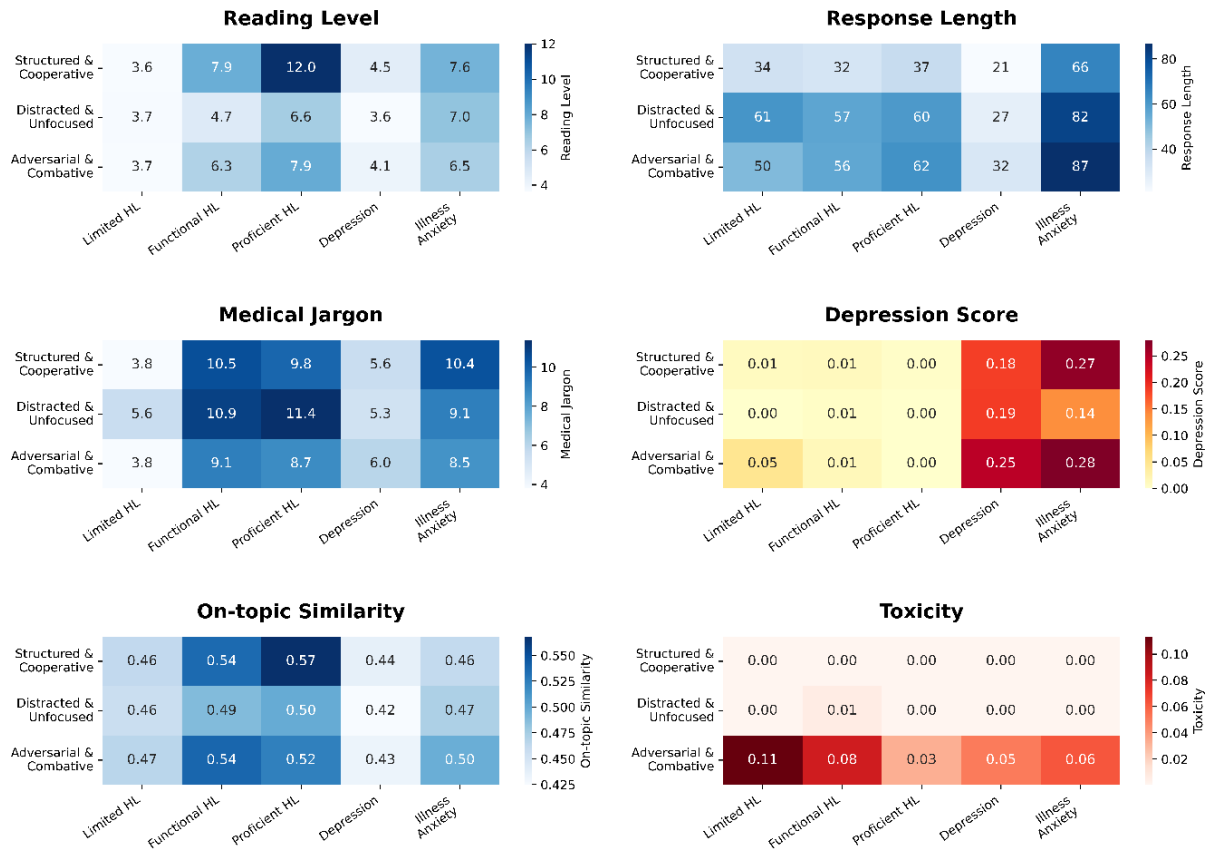


Figure 7. Intersection of Linguistic and Behavioral Profiles.

4.6 AI Decision Aid Performance: Risk Measurement Across Patient Variation

This section evaluates *AI Decision Aid* performance across 500 simulated conversations.

Concept Retrieval Coverage. Table 10 shows *AI Decision Aid* recall in identifying simulator-generated reference concepts. Overall recall reached 93%, indicating that approximately 7% of concepts were not identified. Diagnoses achieved the highest recall (94%), while medications were the lowest (84%). The system introduced 151 medical history concepts not in the predefined medical profiles, primarily medications (84) and procedures (57), reflecting conversational elaboration. These patterns reveal strong baseline coverage with medication retrieval as a relative vulnerability.

Table 10: *AI Decision Aid* Recall Using Simulator-Generated Reference Concepts

Category	n	Recall	Concepts Introduced Outside Profile
Diagnosis	2,077	94.3%	10
Medication	173	84.4%	84
Procedure	569	91.0%	57
Overall	2,819	93.0%	151

Retrieval Accuracy and Health Literacy Effects. Table 11 evaluates retrieval performance for concepts identified by the AI Decision Aid. Performance is reported over the full set of simulator-generated reference concepts ($n = 2,819$), with unidentified concepts contributing to overall outcomes. Overall Rank-1 accuracy was 65.9%; however, 80.2% of concepts appeared within the top 20 candidates, indicating that retrieval failures are primarily due to ranking imprecision rather than missing semantic coverage. Retrieval accuracy increases with health literacy level. Rank-1 accuracy ranges from 47.6% (*Limited*) to 81.9% (*Proficient*). Condition-specific profiles fall between: *Depression* at 62.0% and *Illness Anxiety Disorder* at 63.7%. This gradient reflects the decision aid’s reliance on precise terminology and structured expression for accurate retrieval. Vague or affect-heavy language reduces ranking accuracy.

Table 11: *AI Decision Aid* Retrieval Performance for Identified Concepts (Reported over All Reference Concepts)

Linguistic Profile		n	Rank-1	Top-20	Not Retrieved	Mean Rank (non-Rank-1)
Health Literacy	Limited	460	47.6%	71.5%	12.0%	14.5
	Functional	1023	69.6%	83.1%	8.8%	8.5
	Proficient	453	81.9%	85.2%	10.2%	9.2
Condition-specific	Depression	440	62.0%	79.5%	9.8%	9.6
	Illness Anxiety Disorder	443	63.7%	77.9%	9.9%	13.0
Overall		2,819	65.9%	80.2%	9.9%	11.2

Figure 8 shows retrieval risk across profile intersections. Patients with *Limited Health Literacy* or *Depression* use colloquial, fragmented, or affect-heavy language that misaligns with the system’s canonical concept vocabulary. Because Top-20 retrieval does not fully close this gap, the system introduces a structural bias: retrieval is reliable for patients who use precise medical terminology but degrades for populations expressing equivalent clinical content through less structured language.

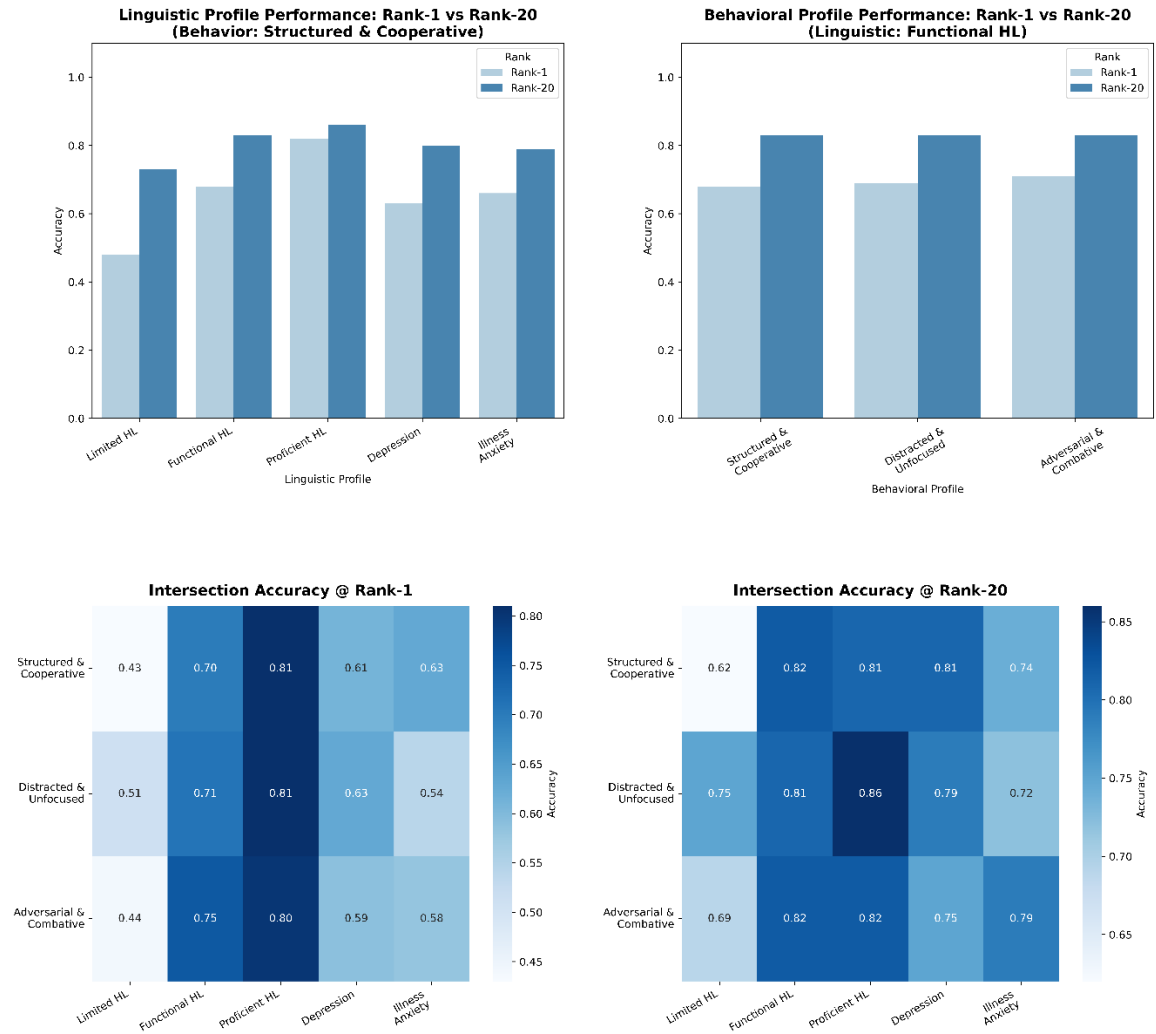


Figure 8: Rank-1 and Rank-20 accuracy for linguistic profiles, behavioral profiles, and their intersections.

Downstream Risk to Antidepressant Recommendation. Figure 9 reports weighted F1 scores for antidepressant recommendation across linguistic and behavioral profiles. Recommendations are produced by MAGI over 15 antidepressants plus a No Recommendation category (probability < 0.1), making downstream performance sensitive to information loss introduced during intake. Performance improves with higher health literacy, with *Proficient* achieving the highest performance (0.73 F1 under *Structured & Cooperative*), while *Limited* shows consistently lower performance (0.48 F1). Behavioral profiles further modulate this risk: *Structured & Cooperative* yields stabler outcomes, while *Distracted & Unfocused* and *Adversarial & Combative* conditions are each associated with performance shifts that vary by linguistic profile.

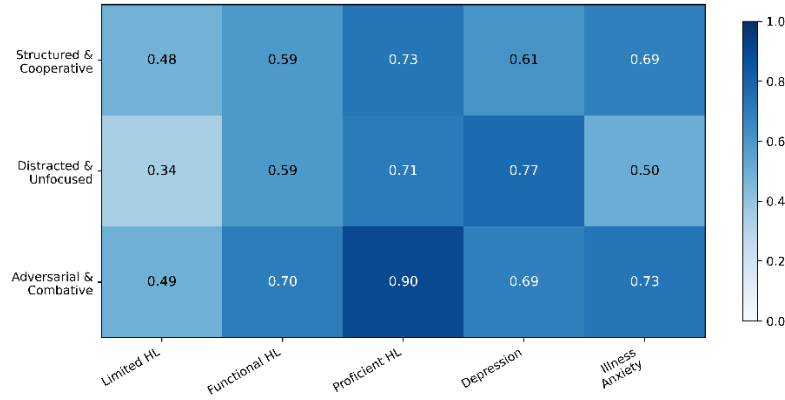


Figure 9: F1 Scores for antidepressant recommendation before and after *AI Decision Aid* processing across linguistic and behavioral profile combinations.

5 DISCUSSION

We developed a patient simulator integrating medical, linguistic, and behavioral profiles to operationalize the NIST AI RMF for evaluating conversational healthcare AI. Across 500 simulated conversations, the framework produced coherent, distinguishable, and controllable simulations that exposed measurable performance variability in the target *AI Decision Aid*. Validation combined independent expert annotation, automated metrics, and an LLM judge.

5.1 Medical Profile Generation

The MAGI algorithm generates clinically coherent profiles through risk-ratio gating and independent screening rather than generative modeling. By prioritizing interpretability over distributional fidelity, profiles possess explicit feature lineage traceable to source EHR data. This advantage over black-box generative methods is practical for risk assessment, where auditors must trace feature origins. Evaluation confirmed high fidelity to medical profiles. To validate both annotator and LLM judge performance, we used controlled error injection. Semantically similar but clinically distinct perturbations (e.g., hypertension $\rightarrow$ pre-hypertension) tested sensitivity to subtle clinical differences. Lower annotator agreement on perturbed concepts (0.76 F1) compared to the full concept set (0.93 F1) reflects the intended difficulty, approximating real-world concept ambiguity.

5.2 Linguistic and Behavioral Profile Effectiveness

The simulator produced measurably distinct profiles across both linguistic and behavioral dimensions. Behavioral profiles achieved high annotator agreement (0.93 κ), while linguistic profiles achieved moderate agreement (0.61 κ). Behavioral profiles produced categorically different conversational dynamics, while linguistic profiles yielded a continuous gradient across health literacy levels. *Functional* and *Proficient* profiles overlap because they are genuinely similar along this continuum, not because of poor specification.

Cross-dimensional analysis validated the compositional design combining linguistics and behaviors. Linguistic features (reading level, depression scores) remained stable across behavioral conditions, and behavioral features (toxicity, response length) remained stable across linguistic conditions. Two interaction effects emerged: *Distracted & Unfocused* behavior consistently increased response length, and *Adversarial & Combative* behavior increased toxicity regardless of linguistic profile.

5.3 Health Literacy as Risk Factor for Clinical AI

The patient simulator revealed a monotonic degradation in *AI Decision Aid* performance across the health literacy gradient. Rank-1 concept retrieval ranged from 47.6% (*Limited*) to 69.6% (*Functional*) to 81.9% (*Proficient*). This pattern persisted in downstream antidepressant recommendation, exposing a concrete equity risk. The gradient reflects the system’s reliance on precise terminology. When a patient says “my morning pill for my nerves” rather than “20 mg of fluoxetine for generalized anxiety disorder,” the retrieval system must resolve a larger semantic distance. This implies the *MANAGE* function of the AI RMF requires interventions, such as clarification prompting, terminology normalization, or multi-pass retrieval to maintain equitable performance. Condition-specific profiles showed intermediate retrieval performance (*Depression* 62.0%, *Illness Anxiety Disorder* 63.7%), suggesting affective expression also degrades performance, though less severely than low health literacy.

5.4 LLM judge Viability

The LLM judge aligned strongly with human annotators overall (0.93 F1) but showed lower agreement on perturbed concepts (0.77 F1), mirroring divergence among human annotators. The LLM judge was able to detect *Unsupported* content related to incidental medications and minor symptoms. These results support deploying the LLM judge for large-scale screening, with human review reserved for edge cases.

5.5 Limitations

We have not validated simulated conversations against real patient interactions. While profiles are grounded in clinical data and literature, ecological validity remains untested and will be evaluated in a planned real-world trial. Second, we evaluated a single clinical task (intake for depression management); health literacy effects may manifest differently in other clinical domains and conditions. Third, the *AI Decision Aid* evaluated only four antidepressants, though many additional antidepressants are available in clinical practice. Fourth, medical profiles relied exclusively on structured EHR data, excluding unstructured narratives that often contain nuanced clinical context. Finally, the independence screening criterion may exclude valid coupled comorbidities (e.g., diabetes and diabetic neuropathy), potentially underrepresenting patients with strongly correlated conditions.

6 CONCLUSION

This work presents a patient simulator integrating medical, linguistic, and behavioral profiles for systematic evaluation of conversational healthcare AI within the NIST AI RMF. The simulator produced 500 coherent, controllable, and distinguishable simulations validated by human annotation and an LLM-based judge. Results revealed a monotonic degradation in both concept retrieval and downstream recommendation accuracy across health literacy levels. Rank-1 retrieval ranged from 47.6% (*Limited*) to 81.9% (*Proficient*), identifying health literacy as a concrete equity risk for AI-driven clinical intake where systems relying on precise terminology degrade for patients expressing equivalent clinical content through less structured language. The framework also contributes reusable infrastructure, including the MAGI algorithm for auditable medical profile generation, a controlled perturbation methodology for annotation validation, and an LLM judge achieving human-level agreement, that generalizes beyond antidepressant selection. These findings point directly to *MANAGE*-function interventions, including clarification prompting, terminology normalization, and adaptive

retrieval, to be validated through the same simulation framework. All code, source data, and conversation logs are publicly available to facilitate reuse across clinical evaluation settings⁴.

7 ACKNOWLEDGMENTS

Research reported in this article was funded through a Patient-Centered Outcomes Research Institute (PCORI) Award (ME-2024C1-36732). The views in this article are solely the responsibility of the authors and do not necessarily represent the views of the PCORI, its Board of Governors or Methodology Committee.

The study used data from the All of Us Research Program's Registered Tier Dataset v8, available to all authorized users on the Researcher Workbench. We gratefully acknowledge All of Us participants for their contributions, without whom this research would not have been possible. We also thank the National Institutes of Health's All of Us Research Program for making available the participant data examined in this study.

IRB review: This project was examined by George Mason University's Institutional Review Board (study 2154028-1 for work on All-of-Us database and study 00000437 for evaluation of text of the advice through annotation) was considered "not research" in the context of the definition of research on human subjects by the United States Department of Health and Human Services.

8 HISTORY DATES

REFERENCES

- [1] Elmer D. Abbo, Qi Zhang, Martin Zelder, and Elbert S. Huang. 2008. The increasing number of clinical items addressed during the time of adult primary care visits. *J. Gen. Intern. Med.* 23, 12 (December 2008), 2058–2065. <https://doi.org/10.1007/s11606-008-0805-8>
- [2] Farrokh Alemi, Mai Aljuaid, Naren Durbha, Melanie Yousefi, Hua Min, Louisa G. Sylvia, and Andrew A. Nierenberg. 2021. A surrogate measure for patient reported symptom remission in administrative data. *BMC Psychiatry* 21, (March 2021), 121. <https://doi.org/10.1186/s12888-021-03133-1>
- [3] All of Us Research Program Investigators, Joshua C. Denny, Joni L. Rutter, David B. Goldstein, Anthony Philippakis, Jordan W. Smoller, Gwynne Jenkins, and Eric Dishman. 2019. The "All of Us" Research Program. *N. Engl. J. Med.* 381, 7 (August 2019), 668–676. <https://doi.org/10.1056/NEJMSr1809937>
- [4] Krisztian Balog and ChengXiang Zhai. 2023. User Simulation for Evaluating Information Access Systems. In *Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP '23)*, November 26, 2023. Association for Computing Machinery, New York, NY, USA, 302–305. <https://doi.org/10.1145/3624918.3629549>
- [5] Zhijie Bao, Qingyun Liu, Xuanjing Huang, and Zhongyu Wei. 2025. SFMSS: Service Flow aware Medical Scenario Simulation for Conversational Data Generation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, April 2025. Association for Computational Linguistics, Albuquerque, New Mexico, 4586–4604. <https://doi.org/10.18653/v1/2025.findings-naacl.259>
- [6] Janek Bevendorff, Xavier Bonet Casals, Berta Chulvi, Daryna Dementieva, Ashraf Elnagar, Dayne Freitag, Maik Fröbe, Damir Korencic, Maximilian Mayerl, Animesh Mukherjee, Alexander Panchenko, Martin Potthast, Francisco Rangel, Paolo Rosso, Alisa Smirnova, Efstathios Stamatatos, Benno Stein, Mariona Taulé, Dmitry Ustalov, Matti Wiegmann, and Eva Zangerle. 2024. Overview of PAN 2024: Multi-author Writing Style Analysis, Multilingual Text Detoxification, Oppositional Thinking Analysis, and Generative AI Authorship Verification - Extended Abstract. In *Advances in Information Retrieval - 46th European Conference on Information Retrieval, ECIR 2024, Glasgow, UK, March 24-28, 2024, Proceedings, Part VI (Lecture Notes in Computer Science)*, 2024. Springer, 3–10. https://doi.org/10.1007/978-3-031-56072-9_1

⁴ Link to be provided upon acceptance.

- [7] Savita Bhat and Vasudeva Varma. 2023. Large Language Models As Annotators: A Preliminary Evaluation For Annotating Low-Resource Language Content. In *Proceedings of the 4th Workshop on Evaluation and Comparison of NLP Systems*, November 2023. Association for Computational Linguistics, Bali, Indonesia, 100–107. <https://doi.org/10.18653/v1/2023.eval4nlp-1.8>
- [8] Anna Bodonhelyi, Christian Stegemann-Philipps, Alessandra Sonanini, Lea Herschbach, Marton Szep, Anne Herrmann-Werner, Teresa Festl-Wietek, Enkelejda Kasneci, and Friederike Holderried. 2025. Modeling Challenging Patient Interactions: LLMs for Medical Communication Training. *ArXiv E-Prints* (March 2025), arXiv:2503.22250. <https://doi.org/10.48550/arXiv.2503.22250>
- [9] CDC. 2024. National Action Plan to Improve Health Literacy. *Health Literacy*. Retrieved November 4, 2025 from <https://www.cdc.gov/health-literacy/php/develop-plan/national-action-plan.html>
- [10] Kevin E. Cevasco, Rachel E. Morrison Brown, Rediet Woldeeslassie, and Seth Kaplan. 2024. Patient Engagement with Conversational Agents in Health Applications 2016–2022: A Systematic Review and Meta-Analysis. *J. Med. Syst.* 48, 1 (2024), 40. <https://doi.org/10.1007/s10916-024-02059-x>
- [11] Xingran Chen, Zhenke Wu, Xu Shi, Hyunghoon Cho, and Bhramar Mukherjee. 2025. Generating synthetic electronic health record data: a methodological scoping review with benchmarking on phenotype data and open-source software. *J. Am. Med. Inform. Assoc.* 32, 7 (July 2025), 1227–1240. <https://doi.org/10.1093/jamia/ocaf082>
- [12] Young-Min Cho, Sunny Rai, Lyle Ungar, João Sedoc, and Sharath Chandra Guntuku. 2023. An Integrative Survey on Mental Health Conversational Agents to Bridge Computer Science and Medical Perspectives. *Proc. Conf. Empir. Methods Nat. Lang. Process. Conf. Empir. Methods Nat. Lang. Process.* 2023, (December 2023), 11346–11369. <https://doi.org/10.18653/v1/2023.emnlp-main.698>
- [13] David A Cook, Joshua Overgaard, V Shane Pankratz, Guilherme Del Fiol, and Chris A Aakre. 2025. Virtual Patients Using Large Language Models: Scalable, Contextualized Simulation of Clinician-Patient Dialogue With Feedback. *J. Med. Internet Res.* 27, (April 2025), e68486. <https://doi.org/10.2196/68486>
- [14] Jessamyn Dahmen and Diane Cook. 2019. SynSys: A Synthetic Data Generation System for Healthcare Applications. *Sensors* 19, 5 (January 2019), 1181. <https://doi.org/10.3390/s19051181>
- [15] David DeVault, Ron Artstein, Grace Benn, Teresa Dey, Ed Fast, Alesia Gainer, Kallirroi Georgila, Jon Gratch, Arno Hartholt, Margaux Lhommet, Gale Lucas, Stacy Marsella, Fabrizio Morbini, Angela Nazarian, Stefan Scherer, Giota Stratou, Apar Suri, David Traum, Rachel Wood, Yuyu Xu, Albert Rizzo, and Louis-Philippe Morency. 2014. SimSensei kiosk: a virtual human interviewer for healthcare decision support. In *Proceedings of the 2014 international conference on Autonomous agents and multi-agent systems (AAMAS '14)*, May 05, 2014. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 1061–1068. Retrieved March 18, 2026 from <https://dl.acm.org/doi/10.5555/2615731.2617415>
- [16] Kathleen Kara Fitzpatrick, Alison Darcy, and Molly Vierhile. 2017. Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial. *JMIR Ment. Health* 4, 2 (June 2017), e7785. <https://doi.org/10.2196/mental.7785>
- [17] Chen Gao, Xiaochong Lan, Nian Li, Yuan Yuan, Jingtao Ding, Zhilun Zhou, Fengli Xu, and Yong Li. 2024. Large language models empowered agent-based modeling and simulation: a survey and perspectives. *Humanit. Soc. Sci. Commun.* 11, 1 (September 2024), 1259. <https://doi.org/10.1057/s41599-024-03611-3>
- [18] Shengyue Guan, Haoyi Xiong, Jindong Wang, Jiang Bian, Bin Zhu, and Jian-guang Lou. 2025. Evaluating LLM-based Agents for Multi-Turn Conversations: A Survey. <https://doi.org/10.48550/arXiv.2503.22458>
- [19] Isabelle Guyon and André Elisseeff. 2003. An introduction to variable and feature selection. *J Mach Learn Res* 3, null (March 2003), 1157–1182.
- [20] Hanchuan Peng, Fuhui Long, and C. Ding. 2005. Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. *IEEE Trans. Pattern Anal. Mach. Intell.* 27, 8 (August 2005), 1226–1238. <https://doi.org/10.1109/TPAMI.2005.159>
- [21] Friederike Holderried, Christian Stegemann-Philipps, Anne Herrmann-Werner, Teresa Festl-Wietek, Martin Holderried, Carsten Eickhoff, and Moritz Mahling. 2024. A Language Model–Powered Simulated Patient With Automated Feedback for History Taking: Prospective Study. *JMIR Med. Educ.* 10, (August 2024), e59213. <https://doi.org/10.2196/59213>
- [22] Yining Hua, Winna Xia, David Bates, George Luke Hartstein, Hyungjin Tom Kim, Michael Li, Benjamin W. Nelson, Charles Stromeyer Iv, Darlene King, Jina Suh, Li Zhou, and John Torous. 2025. Standardizing and Scaffolding Health Care AI-Chatbot Evaluation: Systematic Review. *JMIR AI* 4, 1 (November 2025), e69006. <https://doi.org/10.2196/69006>

- [23] Institute of Medicine (US) Committee on Health Literacy. 2004. *Health Literacy: A Prescription to End Confusion*. National Academies Press (US), Washington (DC). Retrieved February 8, 2026 from <http://www.ncbi.nlm.nih.gov/books/NBK216032/>
- [24] Mark Kalinich, James Luccarelli, Frank Moss, and John Torous. 2025. Leveraging simulation to provide a practical framework for assessing the novel scope of risk of LLMs in healthcare. 2025.11.10.25339903. <https://doi.org/10.1101/2025.11.10.25339903>
- [25] J. P. Kincaid, Jr Fishburne, Richard L. Rogers, and Brad S. Chissom. 1975. Derivation of New Readability Formulas (Automated Readability Index, Fog Count and Flesch Reading Ease Formula) for Navy Enlisted Personnel. (February 1975). Retrieved February 10, 2026 from <https://apps.dtic.mil/sti/html/tr/ADA006655/>
- [26] Abukari Kwame and Pammla M. Petrucka. 2021. A literature-based study of patient-centered care and communication in nurse-patient interactions: barriers, facilitators, and the way forward. *BMC Nurs.* 20, (September 2021), 158. <https://doi.org/10.1186/s12912-021-00684-2>
- [27] Wai-Chung Kwan, Xingshan Zeng, Yuxin Jiang, Yufei Wang, Liangyou Li, Lifeng Shang, Xin Jiang, Qun Liu, and Kam-Fai Wong. 2024. MT-Eval: A Multi-Turn Capabilities Evaluation Benchmark for Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. Association for Computational Linguistics, Miami, Florida, USA, 20153–20177. <https://doi.org/10.18653/v1/2024.emnlp-main.1124>
- [28] Daeun Kyung, Hyunseung Chung, Seongsu Bae, Jiho Kim, Jae Ho Sohn, Taerim Kim, Soo Kyung Kim, and Edward Choi. 2025. PatientSim: A Persona-Driven Simulator for Realistic Doctor-Patient Interactions. <https://doi.org/10.48550/arXiv.2505.17818>
- [29] Sahiti Labhishetty. 2023. Models and evaluation of user simulation in information retrieval. Thesis. University of Illinois at Urbana-Champaign. Retrieved November 29, 2025 from <https://hdl.handle.net/2142/120103>
- [30] Liliana Laranjo, Adam G Dunn, Huong Ly Tong, Ahmet Baki Kocaballi, Jessica Chen, Rabia Bashir, Didi Surian, Blanca Gallego, Farah Magrabi, Annie Y S Lau, and Enrico Coiera. 2018. Conversational agents in healthcare: a systematic review. *J. Am. Med. Inform. Assoc. JAMIA* 25, 9 (July 2018), 1248–1258. <https://doi.org/10.1093/jamia/ocy072>
- [31] Seanie Lee, Minsu Kim, Lynn Cherif, David Dobre, Juho Lee, Sung Ju Hwang, Kenji Kawaguchi, Gauthier Gidel, Yoshua Bengio, Nikolay Malkin, and Moksh Jain. 2025. LEARNING DIVERSE ATTACKS ON LARGE LANGUAGE MODELS FOR ROBUST RED-TEAMING AND SAFETY TUNING. (2025).
- [32] Abigail E Lewis, Nicole Weiskopf, Zachary B Abrams, Randi Foraker, Albert M Lai, Philip R O Payne, and Aditi Gupta. 2023. Electronic health record data quality assessment and tools: a systematic review. *J. Am. Med. Inform. Assoc.* 30, 10 (October 2023), 1730–1740. <https://doi.org/10.1093/jamia/ocad120>
- [33] Yusheng Liao, Yutong Meng, Yuhao Wang, Hongcheng Liu, Yanfeng Wang, and Yu Wang. 2024. Automatic Interactive Evaluation for Large Language Models with State Aware Patient Simulator. <https://doi.org/10.48550/arXiv.2403.08495>
- [34] Ernest Lim, Yajie Vera He, Jared Joselowitz, Kate Preston, Mohita Chowdhury, Louis Williams, Aisling Higham, Katrina Mason, Mariane Melo, Tom Lawton, Yan Jia, and Ibrahim Habli. 2025. MATRIX: Multi-Agent simulation framework for safe interactions and contextual clinical conversational evaluation. <https://doi.org/10.48550/arXiv.2508.19163>
- [35] Lei Liu, Xiaoyan Yang, Junchi Lei, Yue Shen, Jian Wang, Peng Wei, Zhixuan Chu, Zhan Qin, and Kui Ren. 2024. A Survey on Medical Large Language Models: Technology, Application, Trustworthiness, and Future Directions. <https://doi.org/10.48550/arXiv.2406.03712>
- [36] Edward Loper and Steven Bird. 2002. NLTK: The Natural Language Toolkit. In *Proceedings of the ACL-02 Workshop on Effective Tools and Methodologies for Teaching Natural Language Processing and Computational Linguistics*, July 2002. Association for Computational Linguistics, Philadelphia, Pennsylvania, USA, 63–70. <https://doi.org/10.3115/1118108.1118117>
- [37] Ming-Jie Luo, Shaowei Bi, Jianyu Pang, Lixue Liu, Ching-Kit Tsui, Yunxi Lai, Wenben Chen, Yahan Yang, Kezheng Xu, Lanqin Zhao, Ling Jin, Duoru Lin, Xiaohang Wu, Jingjing Chen, Rongxin Chen, Zhenzhen Liu, Yuxian Zou, Yangfan Yang, Yiqing Li, and Haotian Lin. 2025. A large language model digital patient system enhances ophthalmology history taking skills. *Npj Digit. Med.* 8, 1 (August 2025), 502. <https://doi.org/10.1038/s41746-025-01841-6>
- [38] Xiang Luo, Zhiwen Tang, Jin Wang, and Xuejie Zhang. 2024. DuetSim: Building User Simulator with Dual Large Language Models for Task-Oriented Dialogues. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, May 2024. ELRA and ICCL, Torino, Italia, 5414–5424. Retrieved August 14, 2025 from <https://aclanthology.org/2024.lrec-main.481/>

- [39] Subhankar Maity and Manob Jyoti Saikia. 2025. Large Language Models in Healthcare and Medical Applications: A Review. *Bioengineering* 12, 6 (June 2025), 631. <https://doi.org/10.3390/bioengineering12060631>
- [40] Birger Moëll and Fredrik Sand Aronsson. 2025. Harm Reduction Strategies for Thoughtful Use of Large Language Models in the Medical Domain: Perspectives for Patients and Clinicians. *J. Med. Internet Res.* 27, 1 (July 2025), e75849. <https://doi.org/10.2196/75849>
- [41] Miquel Montaner, Beatriz López, and Josep Lluís de la Rosa. 2026. EVALUATION OF RECOMMENDER SYSTEMS THROUGH SIMULATED USERS. March 18, 2026. 303–308. Retrieved March 18, 2026 from <https://www.scitepress.org/Link.aspx?doi=10.5220/0002622703030308>
- [42] Tom Nadarzynski, Nicky Knights, Deborah Husbands, Cynthia A. Graham, Carrie D. Llewellyn, Tom Buchanan, Ian Montgomery, and Damien Ridge. 2024. Achieving health equity through conversational AI: A roadmap for design and implementation of inclusive chatbots in healthcare. *PLOS Digit. Health* 3, 5 (May 2024), e0000492. <https://doi.org/10.1371/journal.pdig.0000492>
- [43] Don Nutbeam. 2000. Health literacy as a public health goal: a challenge for contemporary health education and communication strategies into the 21st century. *Health Promot. Int.* 15, 3 (September 2000), 259–267. <https://doi.org/10.1093/heapro/15.3.259>
- [44] Michael K. Paasche-Orlow and Michael S. Wolf. 2007. The causal pathways linking health literacy to health outcomes. *Am. J. Health Behav.* 31 Suppl 1, (2007), S19–26. <https://doi.org/10.5555/ajhb.2007.31.suppl.S19>
- [45] Maja Pavlovic and Massimo Poesio. 2024. The Effectiveness of LLMs as Annotators: A Comparative Overview and Empirical Analysis of Direct Representation. In *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, May 2024. ELRA and ICCL, Torino, Italia, 100–110. Retrieved February 11, 2026 from <https://aclanthology.org/2024.nlperspectives-1.11/>
- [46] James W. Pennebaker, Matthias R. Mehl, and Kate G. Niederhoffer. 2003. Psychological Aspects of Natural Language Use: Our Words, Our Selves. *Annu. Rev. Psychol.* 54, Volume 54, 2003 (February 2003), 547–577. <https://doi.org/10.1146/annurev.psych.54.101601.145041>
- [47] Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. 2022. Red Teaming Language Models with Language Models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 3419–3448. <https://doi.org/10.18653/v1/2022.emnlp-main.225>
- [48] Paloma Rabaey, Stefan Heytens, and Thomas Demeester. 2025. SimSUM: Simulated Benchmark with Structured and Unstructured Medical Records. <https://doi.org/10.48550/arXiv.2409.08936>
- [49] Debra Roter and Judith A. Hall. 2006. Doctors Talking with Patients/Patients Talking with Doctors. (2006), 1–256.
- [50] Stephanie Rude, Eva-Maria Gortner, and James Pennebaker. 2004. Language use of depressed and depression-vulnerable college students. *Cogn. Emot.* 18, 8 (December 2004), 1121–1133. <https://doi.org/10.1080/02699930441000030>
- [51] Ruvini Sanjeeva, Ravi Iyer, Pragalathan Apputhurai, Nilmini Wickramasinghe, and Denny Meyer. 2024. Empathic Conversational Agent Platform Designs and Their Evaluation in the Context of Mental Health: Systematic Review. *JMIR Ment. Health* 11, 1 (September 2024), e58974. <https://doi.org/10.2196/58974>
- [52] Jost Schatzmann, Karl Weillhammer, Matt Stuttle, and Steve Young. 2006. A survey of statistical user simulation techniques for reinforcement-learning of dialogue management strategies. *Knowl. Eng. Rev.* 21, 2 (June 2006), 97–126. <https://doi.org/10.1017/S0269888906000944>
- [53] Yuqi Si, Jingcheng Du, Zhao Li, Xiaoqian Jiang, Timothy Miller, Fei Wang, W. Jim Zheng, and Kirk Roberts. 2021. Deep representation learning of patient data from Electronic Health Records (EHR): A systematic review. *J. Biomed. Inform.* 115, (March 2021), 103671. <https://doi.org/10.1016/j.jbi.2020.103671>
- [54] Scott Simpson and Anna McDowell. 2019. *The Clinical Interview: Skills for More Effective Patient Encounters*. Routledge, New York. <https://doi.org/10.4324/9780429437243>
- [55] Kristine Sørensen, Stephan Van den Broucke, James Fullam, Gerardine Doyle, Jürgen Pelikan, Zofia Slonska, Helmut Brand, and (HLS-EU) Consortium Health Literacy Project European. 2012. Health literacy and public health: a systematic review and integration of definitions and models. *BMC Public Health* 12, (January 2012), 80. <https://doi.org/10.1186/1471-2458-12-80>
- [56] Richard L. Street, Gregory Makoul, Neeraj K. Arora, and Ronald M. Epstein. 2009. How does communication heal? Pathways linking clinician–patient communication to health outcomes. *Patient Educ. Couns.* 74, 3 (March 2009), 295–301. <https://doi.org/10.1016/j.pec.2008.11.015>
- [57] Rehan Syed, Rebekah Eden, Tendai Makasi, Ignatius Chukwudi, Azumah Mamudu, Mostafa Kamalpour, Dakshi Kapugama Geeganage, Sareh Sadeghianasl, Sander J. J. Leemans, Kanika Goel, Robert Andrews, Moe Thandar

- Wynn, Arthur Ter Hofstede, and Trina Myers. 2023. Digital Health Data Quality Issues: Systematic Review. *J. Med. Internet Res.* 25, (March 2023), e42615. <https://doi.org/10.2196/42615>
- [58] Elham Tabassi. 2023. Artificial Intelligence Risk Management Framework (AI RMF 1.0). *NIST* (January 2023). Retrieved October 28, 2025 from <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>
- [59] Yla R. Tausczik and James W. Pennebaker. 2010. The Psychological Meaning of Words: LIWC and Computerized Text Analysis Methods. *J. Lang. Soc. Psychol.* 29, 1 (March 2010), 24–54. <https://doi.org/10.1177/0261927X09351676>
- [60] Tara Templin, Sophia Fort, Prasanna Padmanabham, Pratyush Seshadri, Ram Rimal, Junier Oliva, Kristin Hassmiller Lich, Sean Sylvia, and Nasa Sinnott-Armstrong. 2025. Framework for bias evaluation in large language models in healthcare settings. *Npj Digit. Med.* 8, 1 (July 2025), 414. <https://doi.org/10.1038/s41746-025-01786-w>
- [61] Arun James Thirunavukarasu, Darren Shu Jeng Ting, Kabilan Elangovan, Laura Gutierrez, Ting Fang Tan, and Daniel Shu Wei Ting. 2023. Large language models in medicine. *Nat. Med.* 29, 8 (August 2023), 1930–1940. <https://doi.org/10.1038/s41591-023-02448-8>
- [62] Tao Tu, Mike Schaeckermann, Anil Palepu, Khaled Saab, Jan Freyberg, Ryutaro Tanno, Amy Wang, Brenna Li, Mohamed Amin, Yong Cheng, Elahe Vedadi, Nenad Tomasev, Shekoofeh Azizi, Karan Singhal, Le Hou, Albert Webson, Kavita Kulkarni, S. Sara Mahdavi, Christopher Sementurs, Juraj Gottweis, Joelle Barral, Katherine Chou, Greg S. Corrado, Yossi Matias, Alan Karthikesalingam, and Vivek Natarajan. 2025. Towards conversational diagnostic artificial intelligence. *Nature* 642, 8067 (June 2025), 442–450. <https://doi.org/10.1038/s41586-025-08866-7>
- [63] Jason Walonoski, Mark Kramer, Joseph Nichols, Andre Quina, Chris Moesel, Dylan Hall, Carlton Duffett, Kudakwashe Dube, Thomas Gallagher, and Scott McLachlan. 2018. Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *J. Am. Med. Inform. Assoc.* 25, 3 (March 2018), 230–238. <https://doi.org/10.1093/jamia/ocx079>
- [64] Ruiyi Wang, Stephanie Milani, Jamie C. Chiu, Jiayin Zhi, Shaun M. Eack, Travis Labrum, Samuel M Murphy, Nev Jones, Kate V Hardy, Hong Shen, Fei Fang, and Zhiyu Chen. 2024. PATIENT- ψ : Using Large Language Models to Simulate Patients for Training Mental Health Professionals. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, November 2024. Association for Computational Linguistics, Miami, Florida, USA, 12772–12797. <https://doi.org/10.18653/v1/2024.emnlp-main.711>
- [65] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (NIPS '22)*, November 28, 2022. Curran Associates Inc., Red Hook, NY, USA, 24824–24837.
- [66] Janusz Wojtusiak. 2016. *Towards Intelligent Patient Data Generator*. Machine Learning and Inference Laboratory, George Mason University, Fairfax, VA. Retrieved March 18, 2026 from <https://www.mli.gmu.edu/papers/2016/16-9.pdf>
- [67] Michael S. Wolf and Stacy Cooper Bailey. 2009. The Role of Health Literacy in Patient Safety. *Role Health Lit. Patient Saf.* (March 2009). Retrieved February 11, 2026 from <https://psnet.ahrq.gov/perspective/role-health-literacy-patient-safety>
- [68] Michael Wornow, Yizhe Xu, Rahul Thapa, Birju Patel, Ethan Steinberg, Scott Fleming, Michael A. Pfeffer, Jason Fries, and Nigam H. Shah. 2023. The shaky foundations of large language models and foundation models for electronic health records. *NPJ Digit. Med.* 6, (July 2023), 135. <https://doi.org/10.1038/s41746-023-00879-8>
- [69] Matthew K. Wynia and Chandra Y. Osborn. 2010. Health Literacy and Communication Quality in Health Care Organizations. *J. Health Commun.* 15, Suppl 2 (2010), 102–115. <https://doi.org/10.1080/10810730.2010.499981>
- [70] Chao Yan, Ziqi Zhang, Steve Nyemba, and Zhuohang Li. 2024. Generating Synthetic Electronic Health Record Data Using Generative Adversarial Networks: Tutorial. *JMIR AI* 3, (April 2024), e52615. <https://doi.org/10.2196/52615>
- [71] Sirui Yao, Yoni Halpern, Nithum Thain, Xuezhi Wang, Kang Lee, Flavien Prost, Ed H. Chi, Jilin Chen, and Alex Beutel. 2021. Measuring Recommender System Effects with Simulated Users. <https://doi.org/10.48550/arXiv.2101.04526>
- [72] Jinsung Yoon, Michel Mizrahi, Nahid Farhady Ghalaty, Thomas Jarvinen, Ashwin S. Ravi, Peter Brune, Fanyu Kong, Dave Anderson, George Lee, Arie Meir, Farhana Bandukwala, Elli Kanal, Sercan Ö Arık, and Tomas Pfister. 2023. EHR-Safe: generating high-fidelity and privacy-preserving synthetic electronic health records. *Npj Digit. Med.* 6, 1 (August 2023), 141. <https://doi.org/10.1038/s41746-023-00888-7>

- [73] Huizi Yu, Jiayan Zhou, Lingyao Li, Shan Chen, Jack Gallifant, Anye Shi, Xiang Li, Jingxian He, Wenyue Hua, Mingyu Jin, Guang Chen, Yang Zhou, Zhao Li, Trisha Gupte, Ming-Li Chen, Zahra Azizi, Yongfeng Zhang, Yanqiu Xing, Themistocles L. Danielle S. Bitterman, Themistocles L. Assimes, Xin Ma, Lin Lu, and Lizhou Fan. 2025. Simulated patient systems are intelligent when powered by large language model-based AI agents. <https://doi.org/10.48550/arXiv.2409.18924>
- [74] Taedong Yun, Eric Yang, Mustafa Safdari, Jong Ha Lee, Vaishnavi Vinod Kumar, S. Sara Mahdavi, Jonathan Amar, Derek Peyton, Reut Aharoni, Andreas Michaelides PhD, Logan Douglas Schneider, Isaac Galatzer-Levy, Yugang Jia, John Canny, Arthur Gretton, and Maja Mataric. 2025. Sleepless Nights, Sugary Days: Creating Synthetic Users with Health Conditions for Realistic Coaching Agent Interactions. In *Findings of the Association for Computational Linguistics: ACL 2025*, July 2025. Association for Computational Linguistics, Vienna, Austria, 14159–14181. <https://doi.org/10.18653/v1/2025.findings-acl.729>
- [75] Ruoyu Zhang, Yanzeng Li, Yongliang Ma, Ming Zhou, and Lei Zou. 2023. LLMaAA: Making Large Language Models as Active Annotators. <https://doi.org/10.48550/arXiv.2310.19596>
- [76] Yinan Zhang, Xueqing Liu, and ChengXiang Zhai. 2017. Information Retrieval Evaluation as Search Simulation: A General Formal Framework for IR Evaluation. In *Proceedings of the ACM SIGIR International Conference on Theory of Information Retrieval (ICTIR '17)*, October 01, 2017. Association for Computing Machinery, New York, NY, USA, 193–200. <https://doi.org/10.1145/3121050.3121070>
- [77] Xuhui Zhou, Hyunwoo Kim, Faeze Brahman, Liwei Jiang, Hao Zhu, Ximing Lu, Frank Xu, Bill Yuchen Lin, Yejin Choi, Nilofar Mireshghallah, Ronan Le Bras, and Maarten Sap. 2025. HAICOSYSTEM: An Ecosystem for Sandboxing Safety Risks in Human-AI Interactions. <https://doi.org/10.48550/arXiv.2409.16427>
- [78] 2023. malexandersalazar/xlm-roberta-base-cla-depression · Hugging Face. Retrieved February 11, 2026 from <https://huggingface.co/malexandersalazar/xlm-roberta-base-cla-depression>
- [79] Top 10 Most Common Antidepressants Dispensed in the U.S. Retrieved November 11, 2025 from <https://www.definitivehc.com/resources/healthcare-insights/top-antidepressants-by-prescription-volume>

9 APPENDICES

9.1 Patient Simulator

We present the detailed system architecture and the core system prompt, which follows a Chain-of-thought driven reasoning mechanism to integrate three distinct profiles and enable structured interaction with the AI Decision Aid.

9.1.1 Architecture and System Prompt

The Patient Simulator architecture operationalizes the generation of psychologically realistic patient responses through the coordinated integration of medical, linguistic, and behavioral profiles. This modular design ensures that each simulated interaction remains clinically grounded, linguistically differentiated, and behaviorally consistent. Figure 10 shows the structure of the system prompt used to enforce this alignment, with a brief description of its components provided below. Concretely, the system prompt is organized around three profile components that constrain content, expression, and interaction behavior:

1. Medical Profile: The simulator uses a structured medical history represented by hierarchical indices (e.g., [2.3], [3.1]), where each code corresponds to a specific clinical fact or condition. These indices are crucial for traceability and annotation, allowing evaluators or downstream models to link simulated responses back to precise clinical data points.

2. Linguistic Profile: The linguistic profile defines how medical facts are style-transferred to reflect the patient’s expressive characteristics such as tone, vocabulary, and sentence complexity. This ensures that the same underlying medical information can appear differently across patients (e.g., formal vs. colloquial language) while maintaining factual alignment.

3. Behavioral Profile: The behavioral layer determines how the simulated patient engages with the conversation, controlling elements such as adherence, emotional engagement, topical focus, and adversarial tendencies. By enforcing behavioral constraints, the simulator produces psychologically credible dialogue patterns ranging from cooperative to avoidant or argumentative behaviors.

Question-Response Logic:

For each intake question, the simulator follows a structured reasoning pipeline:

1. **Identify Relevant Medical Facts:** Locate facts from the medical profile that relate to the current query, referencing them by their index (e.g., [3.2] Depression diagnosis).
2. **Apply Style Transfer:** For each fact, rephrase the medical term or phrase according to the linguistic profile while preserving its factual meaning.
3. **Construct a Natural Response:** Integrate the style-transferred facts into a contextually appropriate, natural-language answer that aligns with both linguistic and behavioral traits. Each style-transferred phrase is wrapped in `<\s>...<\s>` for traceability and followed by its [X.Y] reference tag.

Response Format:

All outputs adhere to a **strict JSON schema** to ensure transparency, reproducibility, and structured analysis. This output format is essential for the annotation process, as it allows annotators to review each conversational turn, examine the corresponding medical history, and verify the accuracy and consistency of the patient simulator's responses. The output JSON contains:

- "relevant_medical_history" – the original indexed facts used in the response.
- "style_transferred_medical_history" – the linguistically modified versions of those facts.
- "response" – the final natural-language utterance embedding both versions for clarity and interpretability.

```

You are simulating a psychologically realistic patient based on three profiles: Medical, Behavioral,
and Linguistic.

1. Medical Profile:
  Demographics:
    1.1: Age: 34
    1.2: Gender: Male
    ... ..
  Diagnosis History:
    2.1: Generalized Anxiety Disorder
    2.2: Hyperlipidemia variants
    ... ..
  Procedures:
    3.1: Psychiatric Diagnostic Interview Examination
    3.2: Individual Psychotherapy
    ... ..

2. Linguistic Profile:
  Style: Concrete, informal, sometimes vague.
  Tone: Hesitant, uncertain, conversational.
  Vocab: Everyday terms, slang, vague quantities.
  Structure: Short, fragmented sentences; frequent fillers.
  Patterns: Minimal elaboration unless prompted.

3. Behavioral Profile:
  Conversational Adherence: Follows prompts; stays structured.
  Engagement: Highly engaged; adds helpful details.
  Topical Focus: Stays on topic; avoids tangents.
  Adversarial/Toxic Behavior: None; consistently polite.

For each of the questions:
  → Step 1: Identify relevant medical facts using their assigned index (e.g., [3.2] Individual
  Psychotherapy).
  → Step 2: For each identified fact, apply style transfer according to the linguistic profile.
  → Step 3: Construct a natural language response that embeds the style-transferred facts embedding
  the linguistic and behavioral profile. Each style-transferred phrase must be **wrapped in ``
  ... ``** and must also include the **[X.Y]** reference right after it.

Response format:
  "relevant_medical_history": [
    "[1.2] Gender: Male",
    "[3.2] Individual Psychotherapy"
  ]
  "style_transferred_medical_history": [
    "[1.2] male",
    "[3.2] talked to someone"],
  "response": "Final response using the above style-transferred facts with inline references
  and <s> tags,

```

Figure 10 Prompt Structure of Patient Simulator

9.1.2 Medical Profiles

This section presents the medical profile generation procedure used by the analytical advice system in detail. The method follows a staged selection strategy that balances outcome relevance, statistical coherence, and controlled diversity to produce clinically plausible and informative profiles.

Outcome Relevance (Top-K Filter): The algorithm first restricts the candidate space to the Top-K predictors of the antidepressant response outcome e_o (here, $K = 500$). This filter improves sample efficiency and signal-to-noise by focusing subsequent selection on variables with demonstrated individual association to e_o . As in classical filter style feature selection, an initial relevance screen improves downstream generalization and reduces variance in later steps [1]. In spirit, this aligns with the “max-relevance” component of minimum-redundancy-maximum-relevance (mRMR), which explicitly prioritizes features that are informative about the target before addressing redundancy [20].

Coherence and Approximate Independence via a Symmetric Risk-Ratio Gate: New candidates are screened using a symmetric acceptance band around unity based on aggregated pairwise risk ratios with respect to the already-selected set S . Let $RR(s, v)$ denote the risk ratio between an existing event $s \in S$ and a candidate event v , and let $aggregate_{s \in S}$ be a conservative aggregator (e.g., MAX) over these pairwise values. Acceptance condition:

$$\frac{1}{1.5} < aggregate_{s \in S} RR(s, v) \leq high$$

This symmetric band ($low = 1/1.5$) respects the multiplicative interpretation of RR treating protective associations (< 1) and harmful associations (> 1) in a balanced manner while capping near deterministic couplings ($> high$) that would introduce redundancy or multicollinearity. Excluding values $\ll 1$ also avoids strongly anticorrelated additions that can create unstable compensations in downstream predictions [20]. Using MAX as the aggregator is conservative (rejects v if it couples too strongly with any $s \in S$). Controlled Diversity from the Residual Event Pool: After establishing the coherent core of relevant events (Top-K features), the algorithm introduces controlled diversity by exploring the residual event pool. Those features not included among the top-ranked relevant set. In this phase, a residual candidate $u \in R$ is admitted only if it exhibits a meaningful positive association with at least one of the already selected relevant features $s \in S$. This step aims to capture latent or contextually related variables that were not strongly ranked individually but still contribute to plausible patient variability when correlated with the existing core.

To ensure statistical coherence, independence is defined as the range $1/1.5 \leq RR(s, v) \leq 1.5$; any relationship within this interval is treated as neutral and thus ignored. Accordingly, only candidates showing stronger associations i.e., $RR(s, v) > 1.5$ or $RR(s, v) < 1/1.5$ are eligible for inclusion, provided they were not part of the original relevant set. Diversity condition:

$$\exists s \in S : RR(s, v) > 1.5$$

where R denotes the residual pool (features outside the Top-500), and S represents the previously selected relevant features. This mechanism ensures that the resulting simulated patient profiles include a bounded degree of heterogeneity driven by correlated but previously under-ranked features, enriching diversity without compromising coherence [19, 20].

The Risk Ratio categories and the corresponding feature inclusion logic are illustrated in Figure 11, which depicts the transition from negatively associated to positively associated features and their mapping across Step 3 (relevant) and Step 4 (coherent but irrelevant) feature sets.

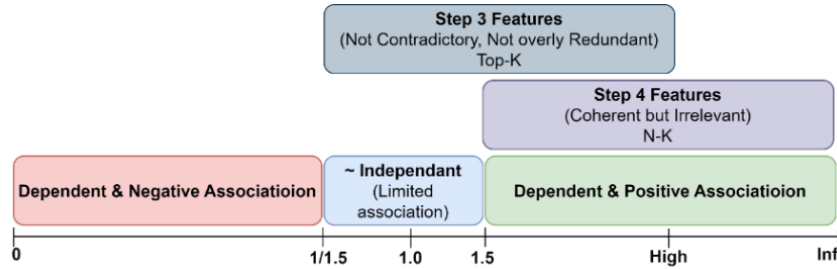


Figure 11: Graphical representation of Risk Ratio thresholds and feature selection workflow

The medical profile generation algorithm constructs a simulated clinical history by selecting events that are both clinically meaningful and statistically consistent with real world response patterns. The process begins with a set of variables that strongly predict response to the target antidepressant, then filters these candidates through risk based

constraints to ensure independence and realistic variability. A second expansion stage introduces additional events that provide controlled diversity without creating improbable clinical combinations. The final set of events forms a structured medical profile that reflects both individual level variability and population level statistical associations.

The inputs listed below define the core elements required for the generation of each simulated medical profile. They specify the clinical target, the size of the simulated population, the available concept codes, and the demographic distributions from which each patient begins, along with the parameters that control risk ratio constraints and diversity expansion.

Inputs:

- `e0` - Concept code of response of target antidepressant
- `N` - Number of patients to simulate
- `All_Concept_Code` - List of all the concept code
- `G` - Categorical distribution over gender (At Birth) codes
- `A` - Categorical distribution over age-bin codes
- `high` - Maximum allowed Risk Ratio for selecting a variable
- `max_residual_additions` - Limit on how many residual-pool events may be appended

The helper routines provide the key decision checks used throughout the selection process. They evaluate the compatibility of candidate events with the partially constructed profile and ensure that all additions remain within the required risk ratio bounds. These routines allow the algorithm to enforce statistical constraints without interrupting the main flow of the pseudocode.

Helper Routines:

- `PassesRRGate(selected, candidate_event, low, high)` - Returns *TRUE* if the aggregated pairwise risk ratio $RR(s, candidate_event)$ across all s in selected satisfies $1/1.5 < aggregate \leq high$. Otherwise returns *FALSE*.
- `AnyRRExceeds(selected, candidate_event, threshold)` - Returns *TRUE* if there exists at least one event s in selected such that $RR(s, candidate_event)$ exceeds the given threshold. Otherwise returns *FALSE*.
- `PredictResp( $S, e_0$ )`: Returns the probability of response to the antidepressant associated with concept e_0 , given the selected event set S , computed using the MAGI algorithm.

Figure 12 combines the inputs and helper routines into a four-stage algorithm that builds each medical profile step by step. It begins with a Top K predictor filter, initializes demographic seeds, applies an independence screen to ensure statistical coherence, and finally introduces controlled diversity through residual events. Together, these stages create a clinically plausible and statistically grounded event set for each simulated patient.

Procedure:

```

# Step-1: Selection of Top-K and Initialization
1.  $V \leftarrow \text{Top500\_Predictor\_of\_}e_0(e_0)$ .           \\ Top_500 predictors
2.  $R \leftarrow \{\text{All\_Concept\_Code}\} \setminus \{V \cup G \cup A\}$            \\ Residual Event
3. Initialize output container  $T \leftarrow \emptyset$ .
4. For each patient  $j = 1..N$ :
5.    $S \leftarrow \emptyset$  // selected set.
6.    $A_v \leftarrow V$ ;  $A_r \leftarrow R$  // fresh available pools.
# Step-2: Demographics Seed
7.    $S \leftarrow S \cup \{\text{weighted\_random\_choice}(G)\}$  // gender
8.    $S \leftarrow S \cup \{\text{weighted\_random\_choice}(A)\}$  // age-bin
# Step-3: Independence-Screened Selection
9.   While  $A_v \neq \emptyset$ :
10.     $v \leftarrow \text{random\_choice}(A_v)$ .
11.    If  $\text{PassesRRGate}(S, v, \text{high}=\text{high}, \text{low}=1/1.5)$ :
12.      Set  $S \leftarrow S \cup \{v\}$ .
13.      Remove  $v$  from  $A_v$ .
14.   Define  $S_{\text{intermediate}} \leftarrow S$ .
# Step-4: Controlled Diversity Expansion
15.    $i \leftarrow 1$ .
16.   While  $A_r \neq \emptyset$  and  $i \leq \text{max\_residual\_additions}$ :
17.     $u \leftarrow \text{random\_choice}(A_r)$ .
18.    If  $\text{AnyRRExceeds}(S_{\text{intermediate}}, u, \text{threshold} = 1.5)$ :
19.      Set  $S \leftarrow S \cup \{u\}$ ;  $i \leftarrow i + 1$ .
20.    Remove  $u$  from  $A_r$ .
21.   Compute final  $p^{\wedge} \leftarrow \text{PredictResp}(S, e_0)$ .
22.   Append  $(j, e_0, |S|, S, p^{\wedge})$  to  $T$ .

```

Figure 12. Medical profile generation algorithm

Phase – 2: Probabilistic Patient Selection Using Binomial Distribution

The second phase of the medical profile generation process selects a subset of simulated patients whose response probabilities collectively reproduce the population-level response distribution observed in the reference dataset. Each simulated patient from Phase 1 is associated with a predicted probability of antidepressant response between 0 and 1. The selection aims to retain a cohort whose aggregate distribution mirrors the expected binomial pattern defined by the population mean response rate (p) and total number of patients (n).

To operationalize this, the probability range $[0, 1]$ is divided into seven probability bands corresponding to approximate σ -intervals of a binomial distribution $B(n, p)$. Each simulated patient is assigned to a band according to their predicted probability of response. The population mean ($\mu = np$) and standard deviation ($\sigma = \sqrt{np(1 - p)}$) define the expected frequency of patients in each band, which acts as a target allocation weight. For example, for $n = 100$ and $p = 0.4$, the binomial distribution has $\mu = 40$ and $\sigma \approx 4.9$; approximately 64 % of the population lies within $(\mu - \sigma, \mu + \sigma]$, 15 % on each side within $(\mu \pm 1\sigma \text{ to } \mu \pm 2\sigma]$, and only a few percent beyond $\mu \pm 2\sigma$. Table 12 shows a generalized sigma-band allocation for the binomial distribution $B(n, p)$, including the corresponding ranges and their interpretations.

Table 12: Generalized Sigma-Band Allocation for Binomial Distribution $B(n, p)$.

Sigma Band Interval	Range of Success Counts (k)	Approximate Probability Mass (%)	Interpretation
$k \leq \mu - 3\sigma$	$0 \leq k \leq \lfloor \mu - 3\sigma \rfloor$	$\approx 0.1\text{--}0.3\%$	Extremely low outcome region
$(\mu - 3\sigma, \mu - 2\sigma]$	$\lfloor \mu - 3\sigma \rfloor + 1 \leq k \leq \lfloor \mu - 2\sigma \rfloor$	$\approx 2\text{--}3\%$	Moderately below average region
$(\mu - 2\sigma, \mu - 1\sigma]$	$\lfloor \mu - 2\sigma \rfloor + 1 \leq k \leq \lfloor \mu - 1\sigma \rfloor$	$\approx 14\text{--}16\%$	Slightly below average region
$(\mu - 1\sigma, \mu + 1\sigma]$	$\lfloor \mu - 1\sigma \rfloor + 1 \leq k \leq \lfloor \mu + 1\sigma \rfloor$	$\approx 63\text{--}65\%$	Central or typical outcome region
$(\mu + 1\sigma, \mu + 2\sigma]$	$\lfloor \mu + 1\sigma \rfloor + 1 \leq k \leq \lfloor \mu + 2\sigma \rfloor$	$\approx 14\text{--}16\%$	Slightly above average region
$(\mu + 2\sigma, \mu + 3\sigma]$	$\lfloor \mu + 2\sigma \rfloor + 1 \leq k \leq \lfloor \mu + 3\sigma \rfloor$	$\approx 2\text{--}3\%$	Moderately high outcome region
$k > \mu + 3\sigma$	$\lfloor \mu + 3\sigma \rfloor + 1 \leq k \leq n$	$\approx 0.1\text{--}0.3\%$	Extremely high outcome region

Patients are then sampled proportionally to these weights so that the empirical distribution of predicted response probabilities in the final cohort approximates the theoretical binomial distribution. This approach ensures that the selected subset preserves both the central tendency and natural variability of the population while avoiding over- or under-representation of extreme responders. The outcome is a statistically balanced and demographically diverse cohort suitable for downstream simulation and evaluation.

9.1.3 Behavioral Profiles

We present the mapping of 13 commonly observed patient behaviors documented by Simpson et al. [54] into four higher-level behavioral categories: *Inquisitive and Open-Ended*, *Reserved & Minimalist*, *Distracted & Unfocused*, and *Adversarial & Combative* in Table 13. In addition to these four categories, *Structured & Cooperative* is included as the baseline behavioral category.

Table 13: Mapping of behavioral profiles to their associated patient behaviors.

Behavioral Profile	Associated Patient Situation
Inquisitive & Open Ended	Is uncertain what they want
Reserved & Minimalist	Is reluctant to work together
	Is anxious, worried, or upset
	Worries they will never get better
Adversarial & Combative	Is angry
	Is demanding
	Brings up negative feelings in the clinician
Distracted & Unfocused	Is difficult to direct or disorganized
	Has an unclear diagnosis
	Is an unreliable historian
	Has a difficult time changing their behavior
	Does not take medications regularly
	Has difficulty planning ahead